

Reserving Capacity, Restoring Fairness: Scheduling Elective Caesarean Sections in the English NHS

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Abstract. Caesarean section is the most common major operation in the English National Health Service (NHS), yet demand for elective caesarean theatre slots routinely exceeds capacity, forcing some patients into emergency pathways, which we term *case losses*, at higher clinical risk and cost. *Early engagers*, whose need is known months before term, and *late engagers*, whose need emerges close to term, share a fixed weekly slot allotment under earliest-deadline-first scheduling. We show that no within-queue scheduling rule can equalize case losses across types, so the managerial lever must move from scheduling to capacity allocation. Using a fluid queueing model with two-sided clinical windows, we derive in closed form the optimal reservation fraction that equalizes a type-specific index balancing case-loss cost against marginal capacity value. The reservation lets every served patient receive a confirmed surgery date at clinical request, eliminating the multi-month wait early engagers currently endure. When the planner also values patient satisfaction from advance confirmation, the optimal reservation shifts further toward early engagers. Calibrated to Cambridge University Hospitals data, the policy reserves 10 of 18 weekly slots for early engagers, cuts late-engager case losses from 22.17% to 9.62%, confirms surgery dates for around 70% of early engagers at clinical request versus zero today, and lowers total case-loss cost by 8.9% (£15,300/year, scaling to over £1.9M if applied across NHS), without additional theatre capacity. The rule operationalizes the RCOG 2025 Good Practice Paper’s call for dedicated elective caesarean operating sessions sized to demand.

Key words: maternity care; elective caesarean sections; capacity reservation; advance confirmation; queueing; healthcare operations

1. Introduction

Caesarean section is the most common major operation performed in the English National Health Service (NHS), with 244,005 caesarean births recorded in England in 2024/25, accounting for 45% of all deliveries (NHS England 2024b). By comparison, the United States recorded a lower aggregate caesarean rate of 32.4% in 2024 and 32.5% in 2025 (Hamilton et al. 2025, 2026). Unlike the US fee-for-service system, where caesarean delivery can be more financially rewarding than vaginal birth, NHS maternity care operates on a bundled-payment system that does not create the same clinician-level incentive for surgical delivery. The English NHS therefore offers a natural setting to study the operational problem of managing elective caesarean demand under fixed theatre capacity, independent of provider financial incentives. At many NHS maternity units, demand for elective caesarean theatre slots routinely exceeds available capacity, leading to safety concerns and poor patient experience (Care Quality Commission 2025). Yet, until recently, there has been little national guidance on how elective caesarean services should be organized. In systems with

hard clinical windows, excess demand inevitably generates *case losses*—patients diverted from the elective pathway to emergency delivery at higher clinical risk and cost—which are the central focus of this paper.

As the Royal College of Obstetricians and Gynaecologists (RCOG) acknowledged in its Good Practice Paper, “prior to this Good Practice Paper there has been no specific guidance pertaining to the organisation of care for women who have a planned caesarean birth” (Royal College of Obstetricians and Gynaecologists 2025). The same guidance calls on hospitals to develop “modelling systems to prospectively identify weeks in which demand for planned caesarean birth is predicted to outstrip supply” and to provide “a pre-determined number of dedicated planned caesarean operating sessions . . . to meet the needs of the population.” This paper operationalizes this guidance by deriving a closed-form capacity-reservation rule for allocating elective caesarean slots across patient types and showing that the same policy substantially increases the number of patients who receive a confirmed surgery date at the time of clinical request.

1.1. The Elective Caesarean Scheduling Problem

All caesarean births in the UK are classified by clinical urgency into four categories (National Institute for Health and Care Excellence 2021a): Category 1 (immediate threat to life; delivery within 30 minutes), Category 2 (maternal or fetal compromise; delivery within 75 minutes), Category 3 (no compromise but early delivery required; no target interval), and Category 4 (timed to suit woman or provider; not before 39 weeks). Categories 1 and 2 are emergency cases managed in dedicated emergency theatres and fall outside the scope of this paper. Following the RCOG 2025 Good Practice Paper, we focus on the *planned* caesarean pathway, also referred to as the *elective* pathway, which in practice includes all Category 4 and clinically stable Category 3 cases managed in planned caesarean theatres.

Within this planned pathway, all patients share a fixed weekly allotment of planned caesarean theatre slots. Each surgery procedure must be performed within a clinically defined two-sided window: not before the earliest gestational age at which surgery is appropriate, and not after the latest date consistent with maternal-fetal safety. This two-sided structure distinguishes elective caesarean scheduling from standard one-sided deadline models; any case not completed within the window must be diverted to the emergency pathway, at higher clinical risk and cost. Patients are classified into two types by when their need for a planned caesarean becomes known to the maternity unit. At our collaborating Cambridge University Hospitals, the operational cutoff between the two patient types is 35 days of advance notice. This classification reflects *information lead time* rather than clinical urgency or patient risk, although it may correlate with indication: the same indication can generate either type depending on when it is identified.

Early engagers are identified well before term as requiring a planned caesarean, typically due to pre-existing indications such as previous caesarean section or maternal preferences, and enter the system around two months before their expected delivery date. Under current practice, however, they typically receive a confirmed surgery date only within a week before delivery, despite early identification, leaving weeks to

months of uncertainty during which they lack a confirmed delivery plan, experience substantial anxiety, and repeatedly contact the scheduling team. This practice deviates from National Institute for Health and Care Excellence (NICE) guidance, which recommends that women requiring a planned caesarean be provided with a confirmed date in advance (National Institute for Health and Care Excellence 2021a). Accordingly, in a patient satisfaction survey at Cambridge University Hospitals, no early engager reported satisfaction with the current scheduling practice. *Late engagers*, by contrast, are identified closer to term because of later-arising or later-detected conditions such as breech presentation at a late scan, unstable lie, late-onset pre-eclampsia, or other deteriorating conditions warranting Category 3-level urgency within the planned pathway. Because their deadlines are imminent upon request, late engagers are typically allocated surgery slots almost immediately when capacity is available and are generally satisfied with their scheduling experience.

The resulting scheduling tension is rooted in earliest-deadline-first (EDF) scheduling—the default protocol in NHS maternity units—which prioritizes patients with the nearest delivery deadlines. Late engagers, whose deadlines are imminent, therefore systematically displace early engagers, whose deadlines are farther away. As a result, early engagement provides no advantage in avoiding case loss. Case losses can be severe: the 2023 MBRRACE-UK report documented a maternal fatality following an out-of-hours Category 3 caesarean, attributed to labour-ward overload and the absence of a dedicated elective operating list (Knight et al. 2023). More broadly, a survey of UK obstetric anaesthetists found that nearly one-quarter of maternity units lack dedicated planned caesarean theatre schedules (Bamber et al. 2023). These failures are structural, not procedural: in an overloaded work-conserving queue, sequencing rules affect which patients receive available slots, but total throughput is pinned down by system capacity (see, e.g., Cachon and Terwiesch 2019). Thus, any within-queue rule that reduces case losses for one patient type must shift the burden to the other. The question is therefore not whether case losses occur, but who bears them and at what cost. Resolving this tension requires capacity allocation, not a different ordering rule. This is precisely the intervention called for by the RCOG guidance: dedicated planned caesarean operating sessions sized to match demand.

1.2. Contributions

Implementing this guidance raises two operational questions. First, how should a maternity unit reserve its weekly planned caesarean capacity between early and late engagers to reflect the relative cost of case loss in each group? Second, can the same reservation policy enable early engagers served through the reserved capacity to receive a confirmed surgery at request, thereby eliminating the prolonged pre-confirmation wait?

To answer these questions, we develop a fluid queueing model of an overloaded planned caesarean system in which two patient types, distinguished by information lead time, share fixed theatre capacity under EDF scheduling. Each case must be served within a two-sided clinical window bounded below by the earliest appropriate gestational age and above by the latest date consistent with maternal-fetal safety. Cases not served within this window are diverted to the costlier emergency pathway. We make three contributions.

First, we provide a structural diagnosis of inequity in current practice. We prove that, in any overloaded non-idling queueing system in which patients face two-sided clinical windows, steady-state case losses for both patient types are characterized by a common service-time threshold determined solely by workload mix and system capacity, independent of the queueing rule and information lead time distributions. Consequently, no within-queue scheduling policy can reallocate case losses between early and late engagers.

Second, we propose a capacity-reservation policy that partitions theatre slots between patient types and derive the optimal split in closed form. At an interior optimum, the policy equalizes a type-specific index balancing case-loss cost against the marginal value of reserved capacity. Because the optimal split is independent of information-lead-time and clinical-window distributions, every early engager served through the reserved capacity can receive a confirmed surgery date at request without altering capacity reservation or case-loss outcomes. We then extend the model to incorporate the value of advance confirmation for early engagers and show that any positive valuation shifts the optimum toward greater reservation for this group.

Third, calibrated to Cambridge University Hospitals data, the optimal policy reserves roughly 10 of 18 weekly theatre slots for early engagers. It reduces late-engager case losses from 22.17% to 9.62%, a 57% relative reduction that more than halves emergency conversion for this clinically vulnerable group. The policy also provides immediate surgery-date confirmation at the time of request to the approximately 10 early engagers per week (about 543 annually) served through the reserved capacity, compared with none under current practice, with an average of 11 weeks of advance notice. Total case-loss cost falls by 8.9%, or around £15,300 annually; scaled to the around 150,000 planned caesareans performed each year in the English NHS—roughly 60% of the 244,005 caesarean births, with the remainder performed as emergency—the same proportional saving amounts to around £1.9 million annually. These gains come with an increase in early-engager case losses from 19.48% to 29.03%—approximately 1.4 additional early engagers per week (74 annually) converting to the emergency pathway. This is a clinically informed trade-off: early-engager losses more often permit a Category 3 or semi-elective extension, whereas late-engager losses more often force Category 1 or 2 emergency conversion. Robustness checks show that the policy remains effective under variation in cost structure, patient mix, no-show rates, time-varying demand, and emergency spillover.

Although calibrated to elective caesarean scheduling, the model applies more broadly to systems in which a capacity-constrained shared resource serves requests with two-sided service windows, heterogeneous arrivals in information lead time, and equity concerns across groups. Examples include deceased-donor organ transplantation and specialist outpatient pathways under statutory wait-time targets, where the same reservation logic and early-confirmation result apply.

2. Literature Review

Our work connects to three research streams: healthcare scheduling, deadline-driven queueing, and pooling versus dedication. We review each in turn below.

2.1. Healthcare Scheduling

Healthcare scheduling has developed a rich set of state-dependent policies grounded in dynamic programming and approximate dynamic programming. Green et al. (2006) formulate a finite-horizon dynamic program for a diagnostic facility serving outpatients, inpatients, and emergency patients, deriving a switching-curve priority rule for real-time service allocation. Patrick et al. (2008) study a dynamic multipriority diagnostic-appointment problem via approximate dynamic programming, assigning future appointment dates subject to priority-specific waiting-time targets. Relatedly, Liu et al. (2010) analyze outpatient appointment scheduling with delay-dependent no-shows and cancellations, dynamically assigning appointment dates while accounting for the probability that scheduled demand fails to materialize. In operating-room settings, Gerchak et al. (1996) derive a reservation rule for elective surgical admissions under uncertain emergency demand, and Huh et al. (2013) extend the elective-versus-emergency allocation problem to dynamic, nonstationary, multi-resource systems with reneging. Broader reviews are provided by Gupta and Denton (2008) for outpatient scheduling and Cardoen et al. (2010) for operating-room planning.

We depart from this literature in three ways, each motivated by the elective caesarean setting. First, early and late engagers differ in when their need becomes known, rather than in *ex ante* clinical priority, so the standard high-versus-low-priority and elective-versus-emergency frameworks do not apply. Second, dynamic-program policies that would override gestational urgency, consultant judgment, or established EDF sequencing are high-dimensional, opaque, and difficult to implement, audit, or defend in maternity operations. We therefore take EDF within each patient group as given and locate the managerial lever upstream, in a transparent capacity-reservation template. Third, our objective is explicitly distributional. Fairness across patient types enters the optimization directly through a tolerance parameter and is jointly balanced against case-loss cost and the patient-experience value of advance surgery-date confirmation. To our knowledge, no prior healthcare-scheduling paper has integrated these considerations in closed form. This feature also distinguishes our work from the canonical price-of-fairness framework of Bertsimas et al. (2011), which studies the efficiency loss from imposing equity constraints on an otherwise efficient allocation. In our setting, total throughput is fixed by capacity, so the central question is not how much efficiency must be sacrificed for equity, but how limited capacity should be partitioned to achieve equity across patient types.

2.2. Deadline-Driven Queueing

Our analysis builds on the deadline-driven queueing literature, where EDF scheduling is a canonical policy but remains analytically challenging. EDF, which serves the job with the most imminent deadline, was introduced by Liu and Layland (1973) in real-time processor scheduling. Panwar et al. (1988) show that non-preemptive EDF stochastically minimizes the fraction of jobs missing their deadlines within the class of non-idling, service-time-independent policies. Tractable characterizations are difficult because the state of an EDF system is not summarized by queue length alone; it must track the distribution of remaining lead

times. Lehoczy (1996) initiate a real-time queueing framework using heavy-traffic approximations for EDF queues, and Doytchinov et al. (2001) extend this approach to $G/G/1$ queues by deriving diffusion limits for the lead-time profile under EDF.

A complementary strand uses fluid (law-of-large-numbers) limits to obtain deterministic characterizations of deadline-driven systems under overload. Atar et al. (2015) establish fluid limits for non-preemptive EDF in $G/G/1+G$ queues with deadlines on service start, showing that stochastic dynamics converge to a deterministic threshold rule as the system grows large. Using a related fluid approach, Ferragut et al. (2025) prove that, in queues where jobs may depart without completing service, the steady-state attained-service distribution is identical under EDF, least-attained-service, and last-come, first-served disciplines. We adopt this methodology to characterize case losses by patient type and establish a parallel invariance insight: in an overloaded queue in which patients face two-sided clinical windows, heterogeneous information lead times create a structural inequity that cannot be eliminated by changing the sequencing rule, because the type-specific case-loss profile is governed by workload mix and system capacity rather than by queue order.

2.3. Pooling versus Dedication

Our policy relates to the literature on pooling versus dedication in service operations. The classical result is that pooling parallel resources improves efficiency through risk pooling and economies of scale when customers and servers are homogeneous. When customer classes differ in urgency, service value, or timing constraints, however, the benefits of pooling may diminish because a single shared allocation rule cannot fully internalize class-specific priorities. For example, Ata and Van Mieghem (2009) show that dedicated class-specific capacity can outperform full pooling when cross-class heterogeneity is sufficiently large.

Our setting extends this logic to elective caesarean scheduling along two clinically important dimensions. First, case-loss costs differ across patient types, yet a pooled EDF queue operating under overload is characterized by a common service-time threshold that does not differentiate between these costs. Second, information lead times differ systematically across patient types, but EDF progressively erodes the head start of an early engager as her time to deadline declines. As a result, pooling fails to translate earlier information into better operational outcomes for early engagers. Another distinction is that traditional pooling-versus-dedication literature focuses primarily on efficiency, whereas our setting also introduces fairness.

3. The Model

Our model represents the planned-caesarean pathway as a maternity unit with a fixed weekly allotment of theatre slots, serving patients whose surgeries must be completed within clinically defined two-sided windows. Each patient's window is bounded below by the earliest gestational age at which surgery is appropriate and above by a hard deadline beyond which delivery becomes clinically unsafe. Patients are heterogeneous in two dimensions: how far in advance the unit learns of their need, and the complexity (service time) of the surgery they require. When demand exceeds capacity, not all patients can be served

within their windows, and scheduling decisions determine which patients are served and which are diverted to the emergency pathway. This creates a trade-off between serving patients already known to the unit and reserving capacity for those whose need will surface later.

Each patient i is characterized by three exogenous calendar dates: a *request date* r_i , an *earliest clinically appropriate surgery date* e_i , and a *deadline date* d_i . The request date r_i is the date on which the decision to pursue a caesarean section is made, either initiated by the patient or recommended by a provider based on clinical history. The dates e_i and d_i delimit a *two-sided clinical window* during which planned caesarean birth is medically feasible: e_i is the earliest date consistent with the patient's indication and gestational age (NICE recommends, for example, surgery at or after 39 weeks unless an earlier date is indicated by neonatal respiratory risk (National Institute for Health and Care Excellence 2021a)), and d_i is the latest delivery date consistent with maternal-fetal safety, set by current obstetric guidance (Royal College of Obstetricians and Gynaecologists 2025). The hospital records each patient in an administrative *booking system* from her request date r_i onward, but the modeled object in our analysis is the *active surgery queue*, which admits patient i only at her *effective eligibility date* $\max\{r_i, e_i\}$; the pre-eligibility interval $[r_i, e_i)$ is an administrative waiting area that the active surgery queue does not see. We define patient i 's *clinical window* T_i , the scheduling horizon seen by the active surgery queue, and *information lead time* I_i , how far in advance of clinical eligibility the booking system learns of her case, by

$$T_i := d_i - \max\{r_i, e_i\}, \quad I_i := e_i - r_i.$$

Patients are classified into two types: *early engagers*, whose information lead time I_i exceeds a clinically chosen threshold, and *late engagers*, whose I_i is at or below the threshold. (At our collaborating Cambridge University Hospitals, the operational threshold is 35 days.) The early/late labels are an information-revelation construct, not a clinical-urgency or risk-severity ordering: the class index tells the planner only how far in advance a patient's need for surgery is known, not how time-critical her case is or how high her clinical risk is. The same indication—e.g., fetal causes—can therefore produce either type depending on detection time. The two types do, however, differ in their conditional indication mix. Late engagers are over-represented in indications whose underlying clinical condition is unstable (e.g., placenta praevia detected at the late scan, late-onset pre-eclampsia, breech presentation) so that slipping past the deadline d_i more often forces a Category 1 or 2 emergency conversion. Early engagers are over-represented in indications with more elastic post-deadline tolerance (e.g., previous caesarean, maternal preference, a history of traumatic delivery) where a deadline slip more often admits a Category 3 or semi-elective extension.

Given this dual heterogeneity in information lead time and post-deadline clinical complexity, the system must decide how to allocate a limited number of operating theatre slots among the two patient types. A *scheduling policy* is a rule that determines when to assign a slot to a patient on the waitlist and which slot

to assign. Given a scheduling policy, each patient who is scheduled receives a *surgery date* s_i —the calendar date her surgery occurs—with $\max\{r_i, e_i\} \leq s_i \leq d_i$. The patient is informed of her assigned surgery date s_i on a *confirmation date* b_i , with $r_i \leq b_i \leq s_i$; the choice of b_i within this interval is governed by a separate *confirmation rule* and does not feed back into the active surgery queue.

A first natural way to handle confirmation is to commit a surgery date at the moment of clinical request, on a first-come, first-served basis by request date r_i . Under such a rule, early engagers—whose request dates precede their clinical windows by months—would lock up future-week capacity before late engagers become known to the system, leaving the late engagers to find the relevant future weeks already full and be diverted to an emergency pathway. We refer to such a diversion of an elective patient to an emergency route as a *case loss*, which is costly in both clinical and operational terms. To avoid this crowd-out, current NHS practice defers confirmation until the patient is close to her clinical window and serves the active surgery queue under *earliest-deadline-first (EDF)*: when a server becomes available, the eligible patient with the smallest remaining time-to-deadline is served (ties broken arbitrarily). EDF reflects the clinical imperative to prioritize the most time-critical case (National Institute for Health and Care Excellence 2021a). The essential consequence is that no patient receives a confirmed slot until she is close to her clinical window.

This paper models the baseline system as a single pooled queue served under EDF. For early engagers, r_i precedes b_i by several weeks to months, creating a long pre-confirmation wait during which the patient has no confirmed surgery date; for late engagers, r_i and b_i are close together; see Figure 1 for an illustration.

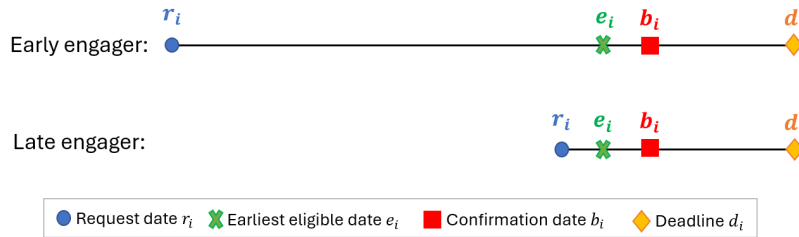


Figure 1 Four dates for each patient type under EDF scheduling: request date r_i , earliest clinically appropriate date e_i , confirmation date b_i , and deadline d_i .

Model primitives. We now formalize the system described above using a stochastic queueing model. In addition to the clinical window T_i already introduced, each patient requires a random amount of surgery time, the *service time* S_i .

Patients request a caesarean according to a Poisson process with total rate λ . Let $\alpha \in [0, 1]$ denote the fraction of early engagers, so the type-specific request rates are $\lambda^E = \alpha\lambda$ and $\lambda^L = (1 - \alpha)\lambda$. We write (S^E, T^E) and (S^L, T^L) for the service-time and clinical-window pairs of early and late engagers, respectively. The deadline d_i refers to the latest date by which surgery must be *completed*, not merely started; accordingly, the clinical window T_i includes the time required for service. (Some papers in the deadline-scheduling literature,

e.g., Lehoczký 1996, define the deadline as the latest permissible start time; our convention aligns with the clinical interpretation in which the patient must be delivered by d_i .) We restrict attention to patients whose request is feasible at the earliest eligible date, requiring $T_i \geq S_i$ almost surely; if $T_i < S_i$, the surgery cannot be completed before the deadline even with an immediate start, and the patient would not enter the system. We use σ and τ as generic coordinates for service time and clinical window, and define the feasible region

$$\mathcal{R}_0 = \{(\sigma, \tau) \in \mathbb{R}^2 \mid \tau \geq \sigma > 0\}. \quad (1)$$

The system has N identical servers (operating theatres), each processing one patient at a time at unit rate. A patient with service time S_i occupies a server for S_i time units. (In practice, theatres operate in discrete scheduling blocks and allocation decisions are made in periodic batches; the continuous-time formulation is a standard queueing-theoretic idealization that preserves the key features of the system—non-idling capacity utilization and EDF priority ordering—while permitting analytical treatment.) Service is non-preemptive, meaning once a patient's surgery begins, it continues uninterrupted until completion.

For each type $k \in \{E, L\}$, let $f^k(\sigma, \tau)$ denote the joint density of (S^k, T^k) on $\mathcal{R}_0$, with service-time marginal

$$f_S^k(\sigma) := \int_{\sigma}^{\infty} f^k(\sigma, x) dx, \quad \sigma > 0,$$

and clinical-window marginal

$$f_T^k(\tau) := \int_0^{\tau} f^k(x, \tau) dx, \quad \tau > 0.$$

The pooled entering population is a mixture of early and late engagers,

$$f(\sigma, \tau) := \alpha f^E(\sigma, \tau) + (1 - \alpha) f^L(\sigma, \tau), \quad (\sigma, \tau) \in \mathcal{R}_0,$$

with pooled marginals $f_S(\sigma) := \alpha f_S^E(\sigma) + (1 - \alpha) f_S^L(\sigma)$ and $f_T(\tau) := \alpha f_T^E(\tau) + (1 - \alpha) f_T^L(\tau)$.

We impose the following regularity and overload conditions.

ASSUMPTION 1.

- (1) **(Bounded and continuous service times)** For each type $k \in \{E, L\}$, S^k has a continuous distribution on $[s_{\min}^k, s_{\max}^k]$ with density f_S^k satisfying $f_S^k(\sigma) > 0$ for all $\sigma \in (s_{\min}^k, s_{\max}^k)$, where $0 < s_{\min}^k \leq s_{\max}^k < \infty$. Furthermore, let $S := \alpha S^E + (1 - \alpha) S^L$; then S has a continuous distribution on $[s_{\min}, s_{\max}]$ with density f_S satisfying $f_S(\sigma) > 0$ for all $\sigma \in (s_{\min}, s_{\max})$.
- (2) **(Bounded and continuous clinical windows with full support)** For each type $k \in \{E, L\}$, T^k has a continuous distribution on $[T_{\min}^k, T_{\max}^k]$ with density f_T^k satisfying $f_T^k(\tau) > 0$ for all $\tau \in (T_{\min}^k, T_{\max}^k)$, where $s_{\min}^k \leq T_{\min}^k < T_{\max}^k < \infty$. Furthermore, let $T := \alpha T^E + (1 - \alpha) T^L$; then T has a continuous distribution on $[T_{\min}, T_{\max}]$ with density f_T satisfying $f_T(\tau) > 0$ for all $\tau \in (T_{\min}, T_{\max})$.
- (3) **(No atom on the boundary)** $\Pr(T = S) = 0$.

ASSUMPTION 2 (Overload). *The offered load exceeds capacity, i.e., $\alpha\lambda\mathbb{E}[S^E] + (1 - \alpha)\lambda\mathbb{E}[S^L] > N$.*

Assumption 1 collects mild regularity conditions. Parts (1)–(2) require service times and clinical windows to be bounded and continuously distributed, which is realistic given finite surgical durations and a fixed planning horizon. The full-support condition on f_T over $(T_{\min}, T_{\max})$ ensures that patients may arrive with any clinical window in this range, consistent with clinical indications for caesarean arising throughout pregnancy. Part (3) is a technical condition ruling out ties between a patient’s service time and clinical window; it holds generically under continuous joint distributions of S and T . Assumption 2 places the system in an overloaded regime where total offered workload exceeds capacity. This is the relevant operational setting, as with ample capacity all patients are served and scheduling becomes trivial.

System state. At time t , each patient i in the system is described by a residual pair $(\sigma_i(t), \tau_i(t)) \in \mathcal{R}_0$, where $\sigma_i(t)$ is the remaining service time and $\tau_i(t)$ is the remaining time-to-deadline. Because service is non-preemptive, waiting and in-service patients must be tracked separately. We aggregate the individual residual pairs into counting measures Φ_t^Q and Φ_t^S on $\mathcal{R}_0$, placing a unit mass at $(\sigma_i(t), \tau_i(t))$ for each waiting and each in-service patient, respectively. The number in service satisfies $\Phi_t^S(\mathcal{R}_0) \leq N$, and we write $\Phi_t := (\Phi_t^Q, \Phi_t^S)$ for the full system state. The active surgery queue admits each patient at her *effective eligibility date* $\max\{r_i, e_i\}$; patients in the pre-eligibility interval $[r_i, e_i)$ are tracked administratively by the booking system but do not appear in Φ_t^Q or Φ_t^S . These measures pool both patient types because EDF schedules solely by remaining time-to-deadline and never uses type information; the aggregate state is therefore sufficient to determine the system dynamics. Type-specific performance metrics are recovered by combining the aggregate dynamics with the type-specific distributions f^E and f^L .

While a patient is waiting, her time-to-deadline $\tau_i(t)$ decreases at unit rate while her service requirement $\sigma_i(t)$ remains constant. Once surgery begins, both decrease at unit rate until $\sigma_i(t)$ reaches zero (i.e., successful completion). The patient is feasible as long as $\tau_i(t) \geq \sigma_i(t)$. Once a patient enters service while feasible, the gap $\tau_i(t) - \sigma_i(t)$ remains constant throughout, guaranteeing that the surgery completes before the deadline. A patient who has not entered service by time $T_i - S_i$ after her effective eligibility date becomes infeasible ($\tau_i(t) < \sigma_i(t)$), meaning elective delivery can no longer be completed before the deadline. An infeasible patient exits the system as a case loss, with per-type clinical, operational, and patient-experience consequences that we formalize as case-loss costs in Section 5. Under the EDF rule, when a server becomes available, the feasible patient with the smallest $\tau_i(t)$ is served next (ties broken arbitrarily).

The stochastic system described above is infinite-dimensional and analytically intractable. We develop a tractable fluid approximation in the next section.

4. Fluid Analysis

To analyze the stochastic system of Section 3, we develop a fluid (law-of-large-numbers) approximation. The key idea is to study a sequence of proportionally scaled systems and show that, as the scale grows, the

stochastic dynamics converge to a deterministic limit. The resulting fluid model inherits the structural logic of EDF while yielding a closed-form characterization of steady-state performance.

4.1. Fluid Regime

Consider a sequence of systems indexed by $n \in \mathbb{N}$, in which both demand and capacity scale proportionally with n . Specifically, in the n -th system, early- and late-engager requests arrive to the system at rates $\lambda^{E(n)} = n\lambda^E$ and $\lambda^{L(n)} = n\lambda^L$, respectively, giving a total arrival rate of $\lambda^{(n)} = n\lambda$. The system has $N^{(n)} = nN$ identical servers, each processing at unit rate with non-preemptive service. Upon request, each early engager draws a service time and clinical window from the type-specific joint distribution $(S^E, T^E) \sim f^E$, and each late engager from $(S^L, T^L) \sim f^L$, independently across requests. The fluid-scaled state variables $q_t^{(n)} := n^{-1}\Phi_t^{Q(n)}$ and $s_t^{(n)} := n^{-1}\Phi_t^{S(n)}$ divide the waiting and in-service counting measures by n . As $n \rightarrow \infty$, the law of large numbers drives these per-unit-scale quantities toward deterministic limits.

To obtain a well-defined limiting description, we impose two standard assumptions.

ASSUMPTION 3 (Initial condition). *As $n \rightarrow \infty$, the fluid-scaled initial state $(q_0^{(n)}, s_0^{(n)})$ weakly converges to a well-defined¹ deterministic limit.*

ASSUMPTION 4 (Fluid-limit convergence). *As $n \rightarrow \infty$, the fluid-scaled state $(q^{(n)}, s^{(n)})$ weakly converges to a deterministic limit, denoted by (q, s) .*

Assumption 3 is a mild regularity condition; it holds, for example, when the system starts from a deterministic initial state such as an empty queue. Assumption 4 is a law-of-large-numbers condition; Atar et al. (2015) establish an analogous result for non-preemptive EDF queues in which deadlines refer to service start, and their proof strategy extends to our setting (deadlines on service completion). Establishing convergence is not the focus of this paper, and we defer the formal statement to Appendix EC.1.1.3.

Figure 2 plots the fluid-scaled waiting mass $q_t^{(n)}$ and in-service mass $s_t^{(n)}$ over time for three system sizes ($n = 1, 10, 100$), each averaged over 30 independent simulation runs. Each system reaches a long-run equilibrium—a *steady state*, denoted with a superscript “*” —in which the state variables no longer change over time. Because the fluid model is deterministic, its steady-state values q^* and s^* are constants (the dashed horizontal lines). The main observation is that the stochastic steady states of the finite- n systems (the curves for $t > 40$) approach these constants, with visible fluctuations at $n = 1$ and close tracking already by $n = 10$, indicating that the fluid approximation reliably predicts the steady-state behavior of finite stochastic systems even at moderate scale. We next characterize the fluid steady state, assuming its existence (Assumption 5).

4.2. Threshold Structure and Steady State

The fluid model is represented by state variables (q_t, s_t) , which are finite measures on $\mathcal{R}_0$ that evolve according to deterministic dynamics. The measure q_t records the distribution of waiting requests over residual pairs (σ, τ) , and s_t records in-service requests, with total in-service mass $s_t(\mathcal{R}_0) \leq N$. As in the

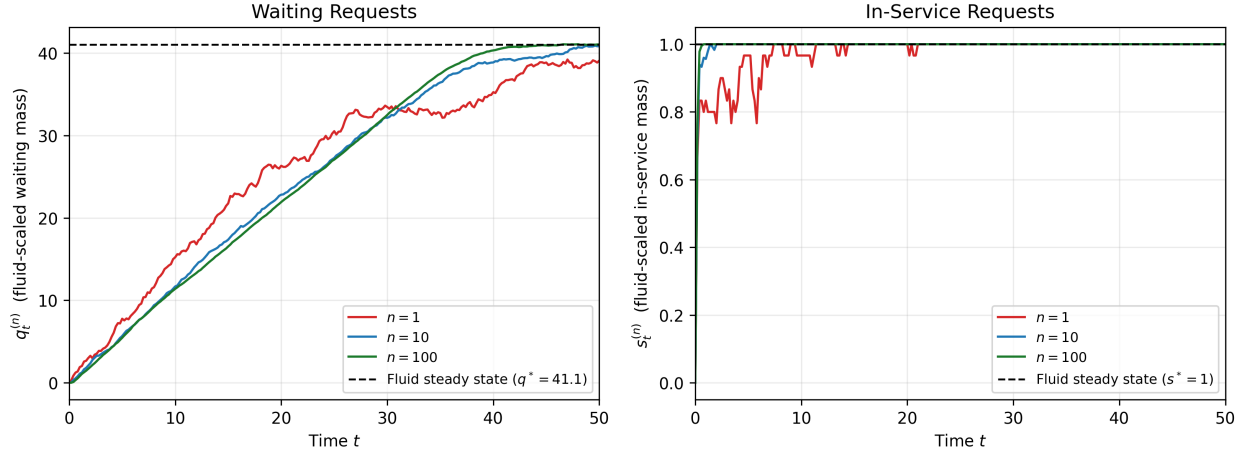


Figure 2 Fluid-scaled state under EDF: simulation vs. fluid approximation.

Note. In the n -th system, $\lambda^{E(n)} = \lambda^{L(n)} = 3n$, $S^E \sim \text{Exp}(5.5)$, $S^L \sim \text{Exp}(4.5)$, $N^{(n)} = n$, $T^E \sim U[5, 15]$, $T^L \sim U[1, 9]$.

Dashed lines represent fluid steady-state values $q^*(\mathcal{R}_0) \approx 41.1$ and $s^*(\mathcal{R}_0) = 1$.

stochastic system, while waiting, a request's remaining time-to-deadline τ decreases at unit rate and its service requirement σ stays fixed; once in service, both τ and σ decrease at unit rate until service completion. The distinction from the stochastic system is that discrete arrivals and discrete admissions to service are replaced by continuous mass flows. Specifically, new mass enters q_t at rate $\lambda f(\sigma, \tau)$, and the transition of waiting mass into in-service mass occurs continuously rather than one request at a time whenever a server becomes free. A waiting request becomes infeasible when $\tau = \sigma$ (recall the feasibility condition (1)), at which point elective caesarean delivery is no longer feasible and the mass exits q_t as a case loss.

The key advantage of the fluid model is that EDF collapses to a single threshold rule. In the finite- n stochastic system, EDF serves the waiting request with the shortest remaining time-to-deadline whenever a server becomes free, but the boundary between served and waiting requests is imprecise: occasional out-of-order admissions occur when a slot opens but no short-deadline request happens to be waiting. As n grows, individual slot openings become a tiny fraction of total capacity. Moreover, since random fluctuations average out, such admissions become negligible relative to the total mass. As a result, the pool of waiting requests with remaining times-to-deadline near the boundary is large enough that EDF can always fill freed capacity with a request whose remaining time-to-deadline is close to the cutoff. In the fluid limit, the separation between served and waiting requests becomes exact: at every time $t > 0$, a single threshold τ_t divides all mass, with requests whose remaining time-to-deadline lies below τ_t in service and those above τ_t waiting.

LEMMA 1 (Threshold separation). *For each $t > 0$ and any Borel set $I \subseteq \mathbb{R}_+$, let $q_t^\tau(I) := q_t(\mathbb{R}_+ \times I)$ and $s_t^\tau(I) := s_t(\mathbb{R}_+ \times I)$ denote the τ -marginals of q_t and s_t . Then there exists a constant $\tau_t > 0$ such that*

$$q_t^\tau([0, \tau_t)) = 0 \quad \text{and} \quad s_t^\tau((\tau_t, \infty)) = 0, \quad \forall t > 0. \quad (2)$$

The separation collapses the two-measure state (q_t, s_t) into a single pooled measure $\Phi_t := q_t + s_t$ plus the scalar τ_t , and reduces the dynamics under EDF to two-region behavior, depicted in the phase diagram of Figure 3. In the waiting region (white area, $\tau \geq \tau_t$), only τ decreases at unit rate while σ stays fixed, so flow moves vertically downward; arrows crossing the horizontal dashed line at $\tau = \tau_t$ represent admission to service. In the service region (blue area, $\tau < \tau_t$), both coordinates decrease at unit rate and flow moves diagonally toward the origin; arrows reaching the vertical axis ($\sigma = 0$) represent service completion. A request whose remaining time-to-deadline drops to σ while still waiting ($\tau = \sigma$ with $\tau > \tau_t$) reaches the feasibility boundary (1): on-time completion is no longer feasible, and since the request cannot enter service until τ falls to $\tau_t < \sigma$, it is lost. In the phase diagram, these losses appear as the downward-moving arrows in the waiting region that cross the diagonal $\tau = \sigma$ above the dashed line.

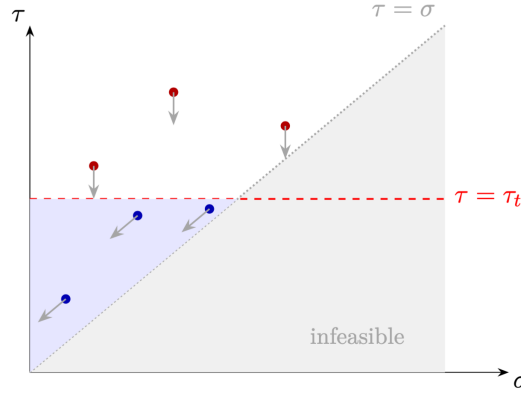


Figure 3 Phase diagram of the EDF fluid model on the feasible region $\mathcal{R}_0$.

Lemma 1 establishes the threshold structure for each t . In the steady state, where $q_t \equiv q^*$, $s_t \equiv s^*$, and $\tau_t \equiv \tau^*$, the dimensionality reduction is especially powerful: instead of solving for the full measure pair (q^*, s^*) on $\mathcal{R}_0$, we need only determine the scalar τ^* . In overload, EDF keeps all servers busy, so the workload admitted per unit time must equal capacity N , pinning τ^* as the unique balance point. Atar et al. (2015) prove steady-state existence for non-preemptive EDF queues in which deadlines refer to service start; we expect the proof to extend to our setting (deadlines on service completion) and assume the conclusion.

ASSUMPTION 5 (Steady-state existence). *The fluid model $(q_t, s_t)_{t \geq 0}$ admits a steady state (q^*, s^*) under non-preemptive EDF.*

Define A^E and A^L as the steady-state case-loss count per unit time for early and late engagers, and let $p^E := A^E / \lambda^E$ and $p^L := A^L / \lambda^L$ denote the corresponding case-loss fractions. Define the *cumulative workload function*

$$h(x) := \int_0^x \sigma f_S(\sigma) d\sigma,$$

which measures the expected workload contributed by arriving requests whose service-time requirements do not exceed x , and its generalized inverse $h^{-1}(y) := \inf\{x \geq 0 : h(x) \geq y\}$.

THEOREM 1 (Steady-state threshold and performance). *Under the EDF policy, there exists a unique threshold $\tau^* \in (s_{\min}, s_{\max})$, where $s_{\min}$ and $s_{\max}$ are the service-time bounds from Assumption 1, satisfying*

$$N = \lambda h(\tau^*). \quad (3)$$

Furthermore,

- (1) *the case-loss fractions are $p^E = \Pr(S^E > \tau^*)$ and $p^L = \Pr(S^L > \tau^*)$, and the case-loss counts per unit time are $A^E = \lambda^E \Pr(S^E > \tau^*)$ and $A^L = \lambda^L \Pr(S^L > \tau^*)$.*
- (2) *the expected waiting times are $\mathbb{E}[(T^E - \max\{S^E, \tau^*\})^+]$ and $\mathbb{E}[(T^L - \max\{S^L, \tau^*\})^+]$.*

Theorem 1 provides the central insight of the fluid analysis. Equation (3) expresses workload balance. Specifically, $h(\tau^*)$ aggregates the arriving workload density $\sigma f_S(\sigma)$ over all service times up to τ^* : $\sigma f_S(\sigma)$ is the fraction of arriving requests with service time in $[\sigma, \sigma + d\sigma]$, and multiplying by σ weights each by the service time it requires. Hence, $\lambda h(\tau^*)$ is the total workload per unit time from requests admitted to service; in steady state, this equals the system capacity N . Equivalently, the threshold is given by $\tau^* = h^{-1}(N/\lambda)$.

Equation (3) is stated in terms of service times, yet EDF schedules by remaining time-to-deadline. The two coincide via the feasibility condition $\tau \geq \sigma$ from (1). Specifically, under EDF, a request enters service when its remaining time-to-deadline τ falls to τ^* , at which point feasibility requires $\sigma \leq \tau^*$. A request with $\sigma > \tau^*$ therefore reaches the case-loss boundary $\tau = \sigma$ before its remaining time-to-deadline can drop to τ^* ; such a request is never served and exits the system as a case loss. Thus, τ^* acts simultaneously as an admission threshold in remaining time-to-deadline and a service-time feasibility cutoff.

The most striking feature of (3) is that it involves only the pooled service-time distribution f_S (which depends on the demand ratio between early and late engagers, namely, λ^E/λ^L), total demand λ , and capacity N . The clinical-window distributions f_T^E and f_T^L are entirely absent. Although EDF is a deadline-based priority rule, the steady-state threshold—and therefore every case-loss fraction—is completely determined by workload and capacity, independent of how far in advance either patient type submits its request. Although the threshold τ^* is common to both patient types, differences in service-time and clinical-window distributions generate type-specific case-loss counts, case-loss fractions, and expected waiting times. The next subsection examines the resulting inequities.

4.3. Structural Inequity

Theorem 1 shows that the current configuration—a single pooled queue operated under EDF—creates a structural inequity: early engagers receive little benefit from early request, while late engagers lose the urgency protection EDF is intended to provide. For early engagers, the disadvantage is twofold. First, EDF schedules patients by remaining time to clinical deadline, not by how long they have been waiting. In the

fluid limit, an early engager is effectively selected only once her remaining time to deadline falls to the threshold τ^* . Thus, an early engager i with request date $r_i \ll d_i - \tau^*$ waits approximately $d_i - r_i - \tau^*$ before receiving a confirmed surgery date. This delay can span weeks or months, during which the patient has no confirmed delivery plan and may experience substantial anxiety. Second, early request does not reduce case-loss risk. The case-loss criterion $\Pr(S^k > \tau^*)$ applies the same service-time cutoff to both patient types, regardless of when the request was placed. Hence, early engagement provides no advantage in avoiding case loss; a patient's outcome is determined by her service-time requirement relative to the pooled threshold.

The same configuration also disadvantages late engagers. EDF is intended to prioritize time-critical cases, but pooling weakens this protection. As early engagers wait, their remaining times to deadline decline steadily. Eventually, some long-waiting early engagers become as deadline-urgent as newly arriving late engagers. EDF may then allocate capacity to an early engager ahead of a late engager whose clinical indications are more complex and whose case-loss cost is higher. In effect, the pooled queue forces late engagers to compete with accumulated early demand, diluting the urgency-based priority that EDF is designed to provide.

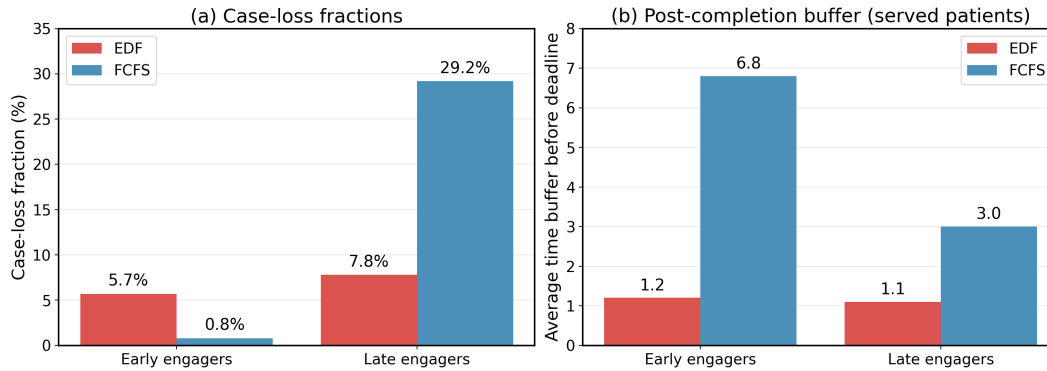


Figure 4 Comparison of EDF and FCFS in the pooled queue.

Note. $\lambda^E = \lambda^L = 3$, $S^E \sim \text{Exp}(5.5)$, $S^L \sim \text{Exp}(4.5)$, $N = 1$, $T^E \sim U[5, 15]$, $T^L \sim U[1, 9]$.

Figure 4 illustrates why changing the sequencing rule within the pooled queue cannot resolve this structural tension. Panel (b) makes the EDF mechanism visible. Under EDF, served patients in both groups complete surgery with little post-completion buffer—the time from surgery completion to the clinical deadline—averaging 1.2 time units for early engagers and 1.1 for late engagers. Thus, although early engagers request much earlier, EDF postpones them until they are nearly as close to deadline as late engagers.

A natural alternative is first-come, first-served (FCFS), which preserves the benefit of early request. Under FCFS, the early-engager case-loss fraction falls from 5.7% to 0.8% (panel (a)), and their average post-completion buffer rises from 1.2 to 6.8 time units (panel (b)). This improvement, however, comes at late engagers' expense: their case-loss fraction rises sharply from 7.8% to 29.2% (panel (a)). Although late engagers who are served under FCFS have a larger average buffer, this reflects selection among who survive

the queue; many more late engagers miss their clinical windows altogether. Thus, within a single overloaded pooled queue, sequencing rules can redistribute case losses across patient groups but cannot eliminate them.

This redistribution is general. Under overload, any non-idling scheduling rule exhausts the available capacity, so any rule that reduces case losses for one type must increase them for the other one-for-one. This sample-path invariance is formalized by Ferragut et al. (2025) for queues in which jobs may depart before completing service. The case-loss trade-off between early and late engagers is therefore a structural consequence of pooled capacity under overload, not an artifact of EDF or any particular scheduling rule.

Since no within-queue sequencing rule can remove this trade-off, the planner must change the queue structure itself. We therefore move from sequencing to capacity reservation: each patient type receives a dedicated share of capacity, while EDF is retained within each type-specific queue. This breaks the pooled-queue invariance that pins τ^* to (f_S, λ, N) and creates type-specific thresholds τ^{E*} and τ^{L*} controlled by a single reservation parameter. The next section derives the optimal reservation rule.

5. Capacity Reservation Policy

This section formalizes and optimizes the capacity-reservation policy motivated by the structural inequity identified in Section 4.3. The reservation acts as a *late-engager protection mechanism*, with capacity dedicated to late engagers preventing displacement by early engagers whose deadlines have become similarly imminent through waiting. We derive the optimal capacity split in closed form, characterize its economic structure, and establish a notable operational implication. Under the reservation policy, early engagers served through the reserved capacity can receive confirmed surgery dates at the time of clinical request—months before delivery—without altering either the capacity reservation or case-loss outcomes.

5.1. Dedicated Queues

Let $\gamma \in [0, 1]$ denote the fraction of servers reserved for early engagers. The reservation policy creates two dedicated queues, one for each type, with respective capacities γN and $(1 - \gamma)N$. Upon request, each patient joins the queue corresponding to her type. Within each queue, patients are scheduled by EDF: when a server becomes available, the eligible patient with the shortest remaining time-to-deadline is served next. As patients wait, their remaining time-to-deadline decreases continuously; if it falls below the required service time, the request becomes a case loss. The system thus operates as two independent single-class EDF queues.

Under reservation fraction γ , let $A^E(\gamma)$ and $A^L(\gamma)$ denote the steady-state case-loss counts per unit time for early and late engagers. To characterize these counts, define the type-specific cumulative workload functions $h^E(x) := \int_0^x \sigma f_S^E(\sigma) d\sigma$ and $h^L(x) := \int_0^x \sigma f_S^L(\sigma) d\sigma$, the type- k analogs of the pooled workload function h from Section 4.2, and their generalized inverses $(h^k)^{-1}$. Define the *critical* reservation fractions

$$\gamma^{max} := \min \left\{ \frac{\lambda^E \mathbb{E}[S^E]}{N}, 1 \right\} \quad \text{and} \quad \gamma^{min} := \max \left\{ 1 - \frac{\lambda^L \mathbb{E}[S^L]}{N}, 0 \right\}.$$

At $\gamma = \gamma^{max}$, the early-engager queue is *critically loaded*, meaning it receives exactly the capacity needed to absorb all early-engager workload. For $\gamma > \gamma^{max}$, the early-engager queue is underloaded and no early engager is lost. Symmetrically, at $\gamma = \gamma^{min}$, the late-engager queue is critically loaded, and for $\gamma < \gamma^{min}$, it is underloaded. Define the *overload interval*

$$I := [\gamma^{min}, \gamma^{max}]. \quad (4)$$

For $\gamma \in I$, both queues are overloaded, so both patient types experience case losses. Outside I , at least one queue is underloaded, and the corresponding patient type experiences no case loss.

5.2. Performance under Reservation

Theorem 1 characterized the pooled queue through the workload-balance equation $N = \lambda h(\tau^*)$, equivalently $\tau^* = h^{-1}(N/\lambda)$. Under the capacity-reservation policy, the system decomposes into two independent single-type EDF queues, and the same balance applies to each queue with type-specific inputs: for patient type $k \in \{E, L\}$, h is replaced by h^k , λ by λ^k , and N by the corresponding reserved capacity (γN or $(1 - \gamma)N$). Unlike the pooled system, however, overload is no longer guaranteed by Assumption 2, since a type-specific queue may become underloaded for $\gamma \notin I$. The following proposition characterizes both regimes.

PROPOSITION 1. *For a given reservation fraction γ , the EDF serving thresholds for early and late engagers are*

$$\tau^{E*}(\gamma) = \begin{cases} (h^E)^{-1}\left(\frac{\gamma N}{\lambda^E}\right), & 0 \leq \gamma < \gamma^{max}, \\ T_{max}^E, & \gamma^{max} \leq \gamma \leq 1, \end{cases}$$

and

$$\tau^{L*}(\gamma) = \begin{cases} T_{max}^L, & 0 \leq \gamma \leq \gamma^{min}, \\ (h^L)^{-1}\left(\frac{(1 - \gamma)N}{\lambda^L}\right), & \gamma^{min} < \gamma \leq 1. \end{cases}$$

The case-loss fractions of early and late engagers are $\Pr(S^E > \tau^{E*}(\gamma))$ and $\Pr(S^L > \tau^{L*}(\gamma))$, respectively, and the case-loss counts per unit time are $A^E(\gamma) = \lambda^E \Pr(S^E > \tau^{E*}(\gamma))$ and $A^L(\gamma) = \lambda^L \Pr(S^L > \tau^{L*}(\gamma))$.

Proposition 1 parallels Theorem 1 but replaces the single pooled threshold τ^* with two type-specific thresholds $\tau^{E*}(\gamma)$ and $\tau^{L*}(\gamma)$, each determined by the corresponding type's service-time distribution and its share of capacity. Overloaded queues have an interior threshold governed by workload balance, while underloaded queues serve all requests. Because the two queues share a fixed total capacity N , the reservation fraction γ pushes the two thresholds in opposite directions. As γ increases, $\tau^{E*}(\gamma)$ rises—admitting more early-engager requests to service and reducing their case-loss fraction—while $\tau^{L*}(\gamma)$ falls, tightening the late-engager queue and increasing their case losses. For $\gamma \in I$, both queues are overloaded and the planner faces a genuine trade-off: improving one type's case-loss fraction necessarily worsens the other's. Outside I , one queue is underloaded and over-served, so the planner can adjust γ to reclaim wasted capacity.

5.3. Optimal Reservation

We now formulate the planner's optimization problem. Formalizing the case-loss costs previewed in Section 3, let $c^E > 0$ and $c^L > 0$ denote the per-case-loss costs for early and late engagers, capturing the clinical, operational, and patient-experience consequences of emergency diversion. The expected total cost rate is

$$C(\gamma) := c^E A^E(\gamma) + c^L A^L(\gamma). \quad (5)$$

The planner seeks a reservation fraction that minimizes total cost while preventing the capacity split from creating a new disparity between the two types. A poorly chosen split could over-correct in either direction, improving one type's case-loss fraction while making the other's unacceptably worse. To guard against this, we impose an equity constraint bounding the gap in case-loss fractions:

$$\left| \frac{A^E(\gamma)}{\lambda^E} - \frac{A^L(\gamma)}{\lambda^L} \right| \leq \Delta,$$

where $\Delta \geq 0$ is the *disparity tolerance*. The constraint does not require equal case-loss fractions—the two types have distinct clinical profiles and may warrant different levels of protection—but it ensures that neither group bears case-loss risk far out of proportion to the other. In the pooled queue, the case-loss gap $|\Pr(S^E > \tau^*) - \Pr(S^L > \tau^*)|$ is fixed by the service-time distributions; the reservation policy now exposes γ as a lever that controls this gap directly. The planner's problem is

$$\min_{\gamma \in [0,1]} C(\gamma) \quad \text{s.t.} \quad \left| \frac{A^E(\gamma)}{\lambda^E} - \frac{A^L(\gamma)}{\lambda^L} \right| \leq \Delta. \quad (6)$$

Before solving (6), we characterize the unconstrained cost minimizer, the minimizer of $C(\gamma)$ over I when the equity constraint is removed.

LEMMA 2 (Unconstrained cost minimizer). *The function $C(\gamma)$ is strictly convex on I , with derivative*

$$C'(\gamma) = N \left(\frac{c^L}{\tau^{L*}(\gamma)} - \frac{c^E}{\tau^{E*}(\gamma)} \right), \quad \gamma \in I. \quad (7)$$

The unique minimizer $\gamma^u := \arg \min_{\gamma \in I} C(\gamma)$ satisfies: if $c^E / \tau^{E}(\gamma) > c^L / \tau^{L*}(\gamma)$ for all $\gamma \in I$, $\gamma^u = \gamma^{max}$; if $c^L / \tau^{L*}(\gamma) > c^E / \tau^{E*}(\gamma)$ for all $\gamma \in I$, $\gamma^u = \gamma^{min}$; otherwise, γ^u is the unique solution in I to*

$$\frac{c^E}{\tau^{E*}(\gamma)} = \frac{c^L}{\tau^{L*}(\gamma)}. \quad (8)$$

The derivative (7) has a direct operational interpretation. For type k , the ratio $c^k / \tau^{k*}(\gamma)$ is the marginal value of capacity in that type's dedicated queue. This value is higher when the per-case-loss cost c^k is larger or when the queue is more congested, as reflected by a smaller threshold τ^{k*} : only patients with shortest service-time requirements can then avoid case loss. Since increasing γ shifts capacity from late engagers to early engagers, $C'(\gamma)$ compares the marginal value of capacity in the late-engager queue with that in the early-engager queue. The interior case of Lemma 2 occurs when these marginal values cross within I . At the

unconstrained optimum γ^u , they are equalized; reallocating capacity in either direction cannot reduce total cost to first order, because the gain in the receiving queue is exactly offset by the loss in the donor queue.

The cost ordering $c^L \geq c^E$ reflects the asymmetric indication mix discussed in Section 3. Late engagers are over-represented among unstable indications, for which missed deadlines more often trigger Category 1 or 2 emergency conversion, whereas early engagers are over-represented among indications with greater post-deadline flexibility. This ordering is an average over the indication mix rather than a claim about every individual indication, and we quantify it in Section 7. Under this ordering, the equalization condition (8) implies $\tau^{L*}(\gamma^u) \geq \tau^{E*}(\gamma^u)$. Thus, at the cost-minimizing reservation level, the late-engager queue receives a weakly more generous service threshold, reflecting the higher expected cost of losing a late engager.

Having characterized the unconstrained minimizer, we now solve the full problem (6) in the next result.

THEOREM 2. *Define the signed fairness gap as*

$$\phi(\gamma) := \frac{A^E(\gamma)}{\lambda^E} - \frac{A^L(\gamma)}{\lambda^L}, \quad \gamma \in I.$$

Then, ϕ is strictly decreasing on I . Let $\underline{\gamma}$ be the unique solution in I to $\phi(\gamma) = \Delta$ (or $\gamma^{\min}$ if no solution exists) and $\bar{\gamma}$ the unique solution to $\phi(\gamma) = -\Delta$ (or $\gamma^{\max}$ if no solution exists). The optimal solution to (6) is

$$\gamma^* = \begin{cases} \underline{\gamma}, & \gamma^u < \underline{\gamma}, \\ \gamma^u, & \underline{\gamma} \leq \gamma^u \leq \bar{\gamma}, \\ \bar{\gamma}, & \gamma^u > \bar{\gamma}. \end{cases}$$

Theorem 2 shows that the optimal policy is the unconstrained cost minimizer γ^u projected onto the equity-feasible interval $[\underline{\gamma}, \bar{\gamma}]$. As γ rises, early-engager case losses fall and late-engager case losses rise, so the signed disparity decreases monotonically from $\phi(\gamma^{\min}) \geq 0$ to $\phi(\gamma^{\max}) \leq 0$. Three regimes arise:

- (i) *Interior optimum* ($\gamma^u \in [\underline{\gamma}, \bar{\gamma}]$). The equity constraint is non-binding: the cost-minimizing split already produces case-loss fractions within the disparity tolerance, and $\gamma^* = \gamma^u$.
- (ii) *Early-engager reservation binds* ($\gamma^u < \underline{\gamma}$). Cost minimization alone would under-reserve for early engagers, producing a case-loss disparity exceeding Δ . The planner increases γ to $\underline{\gamma}$, the smallest reservation fraction that keeps the early-engager case-loss fraction within Δ of the late-engager rate.
- (iii) *Late-engager protection binds* ($\gamma^u > \bar{\gamma}$). Cost minimization alone would over-reserve for early engagers, pushing the late-engager case-loss fraction beyond the disparity tolerance. The planner decreases γ to $\bar{\gamma}$, the largest reservation fraction consistent with equity.

5.4. Early Confirmation of the Surgery Date

Under current practice, schedulers defer the confirmation date b_i for early engagers until a few days before surgery, leaving them without a confirmed delivery plan for weeks to months. By the model of Section 3, however, b_i is set in the booking system and does not feed back into the active surgery queue, which admits patient i only at her effective eligibility date $\max\{r_i, e_i\}$. Choosing b_i is therefore a pure communication

decision; the only operational question is whether the planner knows the surgery date at r_i with enough confidence to commit. The next corollary shows that under the reservation policy, she does.

COROLLARY 1 (Clinical-window and information-lead-time invariance). *The EDF serving thresholds $\tau^{E*}(\gamma)$ and $\tau^{L*}(\gamma)$, the case-loss fractions, the cost function $C(\gamma)$, the fairness gap $\phi(\gamma)$, and the optimal reservation fraction γ^* depend on the service-time distributions (f_S^E, f_S^L) , the arrival rates (λ^E, λ^L) , the capacity N , per-case-loss costs (c^E, c^L) , and the disparity tolerance Δ , but neither on the clinical-window distributions (f_T^E, f_T^L) nor on the information-lead-time distributions (f_I^E, f_I^L) .*

The invariance follows directly from Proposition 1: the threshold $\tau^{E*}(\gamma) = (h^E)^{-1}(\gamma N / \lambda^E)$ depends only on the early-engager service-time distribution and the capacity-to-demand ratio. Neither the clinical-window distribution nor the information-lead-time distribution enters. Two practical implications follow. First, advance confirmation is operationally viable. An early engager's deadline d_i is set by clinical guidance at request and the threshold $\tau^{E*}(\gamma^*)$ is a system constant; hence, the planned surgery date for a served early engager, $s_i \approx d_i - \tau^{E*}(\gamma^*)$, is computable at the request date r_i . The planner can therefore set $b_i = r_i$ for every early engager assigned to reserved capacity. Because b_i never enters the active surgery queue, this advance confirmation leaves case-loss performance and late-engager capacity unchanged; it simply moves the communication of the surgery date from shortly before delivery to clinical request. Second, the optimal reservation γ^* is robust to demand-timing primitives. The planner need not estimate how clinical windows are distributed or how far in advance early engagers will be identified—quantities that vary with referral patterns, patient preferences, and local clinical pathways—so once the service-time distributions, demand, capacity, case-loss costs, and fairness tolerance are calibrated, the same rule transfers across hospitals.

Theorem 2 and Corollary 1 together characterize the complete policy: reserve a fraction γ^* of theatre capacity for early engagers and pin the confirmation date b_i to the request date r_i for every early engager served through the reserved capacity. Late engagers experience little change, since their request date r_i and surgery date s_i are already within days of each other; the experiential transformation therefore concentrates on early engagers, whose pre-confirmation wait would otherwise be months. We next extend the model to incorporate the patient satisfaction value associated with advance confirmation.

6. Early-Engager Satisfaction

Section 5.4 established that the planner can pin $b_i = r_i$ for every early engager served through the reserved capacity without changing case-loss performance. The baseline cost function $C(\gamma)$, however, is insensitive to advance confirmation: it assigns the same value whether some early engagers receive confirmed surgery dates or none do, because case-loss fractions are unchanged (Corollary 1). This blind spot matters: at Cambridge University Hospitals, no early engager reported satisfaction with current scheduling practice, the primary complaint being prolonged uncertainty between request and confirmation. We extend the planner's objective to price this satisfaction and show that any positive valuation, however small, shifts the optimum toward higher early-engager reservation.

6.1. Satisfaction Model

Each confirmed date generates a benefit with at least three observable components: reduced patient anxiety, fewer follow-up contacts with the scheduling team, and the clinical value of a confirmed care plan that enables pre-operative coordination. We capture the combined benefit through a per-slot satisfaction value $v \geq 0$ and define the extended cost function

$$\tilde{C}(\gamma) := c^E A^E(\gamma) + c^L A^L(\gamma) - v\gamma N. \quad (9)$$

The term $v\gamma N$ captures the incremental patient-experience value generated by reserving γN units of capacity for early engagers, thereby enabling surgery-date confirmation at clinical request. This value is assigned only to early engagers because late-engager requests typically occur close to surgery and are already confirmed shortly after request under current practice; reservation therefore adds little advance notice for them. When $v = 0$, $\tilde{C}$ reduces to C , and the results in Section 5 apply unchanged. When $v > 0$, advance confirmation creates value beyond case-loss reduction, strengthening the incentive to reserve capacity for early engagers.

Estimating the monetary value of v is not straightforward, as is estimating c^E and c^L . The extended model is formulated so that the structure of the optimal policy is preserved for any $v \geq 0$: a positive value of advance confirmation shifts the optimal reservation toward early engagers, while its magnitude determines the extent of that shift. We examine the effect of this magnitude through a sensitivity analysis in the Cambridge case study (Section 7.2.2). Once the policy is deployed, observable proxies, such as follow-up contact rates, patient satisfaction scores, and pre-operative pathway efficiency, can support sharper calibration.

6.2. Extended Optimal Reservation

The planner's extended problem replaces $C(\gamma)$ with $\tilde{C}(\gamma)$ in (6):

$$\min_{\gamma \in [0,1]} \tilde{C}(\gamma) \quad \text{s.t.} \quad \left| \frac{A^E(\gamma)}{\lambda^E} - \frac{A^L(\gamma)}{\lambda^L} \right| \leq \Delta. \quad (10)$$

THEOREM 3 (Extended optimal reservation). *The optimal solution $\tilde{\gamma}^*$ to problem (10) is*

$$\tilde{\gamma}^* = \begin{cases} \underline{\gamma}, & \tilde{\gamma}^u < \underline{\gamma}, \\ \tilde{\gamma}^u, & \underline{\gamma} \leq \tilde{\gamma}^u \leq \bar{\gamma}, \\ \bar{\gamma}, & \tilde{\gamma}^u > \bar{\gamma}, \end{cases}$$

where $\tilde{\gamma}^u := \arg \min_{\gamma \in I} \tilde{C}(\gamma)$ is the unique extended unconstrained minimizer over I , and $\underline{\gamma}, \bar{\gamma}$ are as defined in Theorem 2. Moreover, $\tilde{\gamma}^u \geq \gamma^u$, with strict inequality whenever $v > 0$ and γ^u lies in the interior of I .

The satisfaction term $v\gamma N$ is linear in γ , so $\tilde{C}''(\gamma) = C''(\gamma) > 0$ on I , implying that strict convexity is preserved and the extended unconstrained minimizer $\tilde{\gamma}^u$ is unique. Because the equity constraint depends only on case-loss fractions, it is unaffected by v ; hence, the feasible interval $[\underline{\gamma}, \bar{\gamma}]$ and the projection structure of Theorem 2 carry over unchanged.

Analogous to Lemma 2, differentiating the extended cost function in (9) and imposing the first-order condition $\tilde{C}'(\tilde{\gamma}^u) = 0$ at the extended unconstrained minimizer $\tilde{\gamma}^u$ yields

$$\frac{c^L}{\tau^{L*}(\tilde{\gamma}^u)} - \frac{c^E}{\tau^{E*}(\tilde{\gamma}^u)} = v. \quad (11)$$

This condition shows how the satisfaction value modifies the baseline marginal-value balance (8) in Lemma 2. Without satisfaction value ($v = 0$), the unconstrained optimum equalizes the type-specific marginal values of capacity, $c^k/\tau^{k*}(\gamma)$, across $k \in \{E, L\}$, recalling that this ratio is the marginal reduction in case-loss cost per unit of capacity added to queue k . With satisfaction value ($v > 0$), capacity reserved for early engagers generates two benefits: it reduces early-engager case losses and creates satisfaction value v from advance surgery-date confirmation. The planner therefore equalizes augmented marginal values rather than case-loss values alone: the late-engager marginal value of capacity exceeds the early-engager value by exactly v , with the gap offset by the satisfaction generated on the early-engager side.

To compare the extended optimum with the baseline, evaluate $\tilde{C}'$ at γ^u : $\tilde{C}'(\gamma^u) = -vN \leq 0$, so the extended cost is still decreasing in γ at the baseline optimum and the planner should reserve more. By strict convexity of $\tilde{C}$, $\tilde{\gamma}^u \geq \gamma^u$, with strict inequality whenever $v > 0$ and γ^u is interior; any hospital that values patient satisfaction therefore reserves strictly more for early engagers than pure case-loss minimization prescribes. As v grows, $\tilde{\gamma}^u$ shifts further rightward and eventually reaches γ^{max} —the point at which the early-engager queue is critically loaded and every early engager is served—once v exceeds the gap between the late- and early-engager marginal values of capacity at that split. The constrained optimum $\tilde{\gamma}^*$ inherits this shift unless the late-engager protection constraint binds ($\tilde{\gamma}^u > \bar{\gamma}$); in that case $\tilde{\gamma}^* = \bar{\gamma}$, capping the reservation regardless of v so that satisfaction cannot override fairness to late engagers.

The satisfaction extension closes the gap between the baseline model and the hospital's stated concern about patient experience. Even before v is estimated, the direction of the policy shift is unambiguous: reserve more for early engagers. The magnitude, set by (11), is calibrated in the case study, to which we now turn.

7. Case Study: Cambridge University Hospitals

This section calibrates the model to the operational scheduling environment of Cambridge University Hospitals (CUH) and evaluates the baseline policy (pooled EDF) against the capacity-reservation policy (dedicated EDF) via discrete-event simulation. The case study addresses three goals: first, to quantify the inter-class equity gap under pooled EDF and the extent to which capacity reservation reduces aggregate cost while shifting case losses across patient types (Theorem 2); second, to show that even a small valuation of patient satisfaction generates a substantial welfare gain over the pooled baseline (Theorem 3); and third, to assess the robustness of the reservation policy to variation in the cost structure, the early/late classification cutoff, seasonal demand, no-show behavior, and emergency case spillover.

7.1. Data and Parameter Settings

Data. The case study draws on aggregate operational statistics from the 2024 elective caesarean section (ELCS) record of CUH: the annual volume (1,202 elective caesareans), the indication mix, the indication-specific gestational-age windows of clinical eligibility, the request-to-eligibility lead-time distribution, the service-time distribution, and the aggregate case-loss fraction under current practice (namely, pooled EDF).

Parameter calibration. Because patient-level micro-data are not directly available, we use a synthetic generator calibrated to the aggregate statistics to construct an annual stream in which each synthetic patient carries a request date r_i , an indication, a clinical-eligibility window $[e_i, d_i]$, and a service requirement S_i ; the confirmation date b_i is determined by the scheduling policy. The 2024 stream serves as the in-sample case study; a 2025 stream calibrated to the heavier 2025 demand profile (annual volume 1,319) is held out as an out-of-sample cohort for assessing whether the policy gains generalize to a higher-load year. Full specification and reproducibility protocol are documented in Appendix EC.4.2.

We calibrate the generator to the CUH 2024 record along three dimensions. First, the indication mix gives the share of each clinical indication among the 1,202 elective caesareans: previous caesarean (44%), maternal request (20%), breech presentation (9.5%), postdates (3%), fetal causes (7%), maternal medical (6%), with the remaining 10.5% split across smaller indications. Second, the service-time distribution is 96.9% single-slot and 3.1% two-slot in 2024. Third, the indication-specific gestational-age window of clinical eligibility sets the clinical window $T_i = d_i - \max\{r_i, e_i\}$: approximately 7 days for the dominant indications (previous caesarean, maternal request, breech presentation, postdates) and up to 35 days for fetal causes and maternal medical indications. The full per-indication prevalences, gestational-age windows for e_i and d_i , and request-to-eligibility gap distributions are reported in Table EC.5 in Appendix EC.4.2.1.

We then classify patients as early or late engagers by an operational threshold on the information lead time: patient i is an early engager if $I_i = e_i - r_i > 35$ days. This cutoff reflects the distinction between indications identified well before the clinical window, such as previous caesarean or maternal request, and indications that typically emerge close to the clinical window, such as breech presentation, postdates, or unstable lie. Under this threshold, the calibrated generator produces arrival rates of $\lambda^E = 15.00$ and $\lambda^L = 8.11$ patients per week, corresponding to early and late share of 64.9% and 35.1%, respectively, and a total arrival rate of $\lambda = 23.11$ patients per week. Elective caesarean theatre capacity is $N = 18$ slots per week at CUH. The resulting offered load is $\lambda \mathbb{E}[S] = 23.11 \times (0.97 \times 1 + 0.03 \times 2) = 23.11 \times 1.03 = 23.80$ slots per week, yielding a demand-to-supply ratio of 1.32 and placing the system in overload, as required by Assumption 2.

The cost parameters are normalized at $c^E = 1$ and $c^L = 2$, reflecting the clinical judgment that late-engager case losses carry roughly twice the incremental cost of early-engager case losses on the post-deadline trajectory. One normalized unit corresponds to approximately £500 in 2023/24 prices, so $c^E = 1$ and $c^L = 2$ correspond to approximately £500 and £1,000, respectively. This calibration is anchored on the activity-weighted differential between non-elective and elective caesarean unit costs in NHS Reference Costs and is

documented in Appendix EC.4.1.2. The equity tolerance Δ is set to 20% in the baseline; smaller values are reported in the sensitivity analysis. The full set of baseline parameter values is reported in Appendix EC.4.1.

Simulation setup. We compare two scheduling policies on the same synthetic patient stream. Under *pooled EDF*, all patients share a single queue and are served in earliest-deadline-first order, with no class-specific reservation; this is current CUH operational practice. Under *dedicated EDF with capacity reservation*, the elective list is partitioned into two EDF queues with weekly slot allocations $N^E = \gamma^* N$ for early engagers and $N^L = (1 - \gamma^*) N$ for late engagers, where γ^* is the cost-minimizing reservation fraction subject to the equity constraint $|p^L - p^E| \leq \Delta$ from (6). Under both policies, the confirmation date b_i is fixed at the start of the patient's allocated surgery week, mirroring the one-week-ahead booking practice at CUH; this yields a mean confirmation lead time $s_i - b_i$ of approximately 3.5 days in the simulator.

Each scenario is simulated over a horizon of 152 weeks, with the first 100 weeks discarded as warm-up so that all reported statistics are computed from 52 weeks (≈ 1 calendar year) of steady-state operation. Results are averaged over 100 independent replications generated from master seed 42. We employ common random numbers across policies: a single set of synthetic patient traces is generated once and reused across all policies, so that observed differences reflect the scheduling mechanism rather than sampling variability. The full simulation protocol is documented in Appendix EC.4.2.3. The performance metrics of our main focus in this section are the optimal reservation fraction γ^* , the slot allocation (N^E, N^L) , the type-specific case-loss fractions p^E and p^L , the total case-loss cost C , the extended cost $\tilde{C}$ (where applicable), and the cost difference between dedicated EDF and pooled EDF. All cost figures are in normalized units per year. Entries in the result tables are reported as mean $\pm$ SD across the 100 replications.

Model validation. To assess calibration accuracy, we compare simulation outputs under the pooled EDF policy with observed CUH outcomes. On the 2024 in-sample stream, the simulation produces case-loss fractions of $19.48\% \pm 2.14\%$ for early engagers and $22.17\% \pm 2.72\%$ for late engagers (Table 1), with an aggregate case-loss fraction of $20.42\% \pm 2.03\%$. This corresponds to approximately 246.1 ± 31.4 case losses among $1,201.9 \pm 37.5$ patients per year; the CUH 2024 record of 261 case losses among 1,202 patients (21.7%) falls within the simulated ± 1 SD band. To assess out-of-sample performance, we apply the same calibrated generator to the higher-volume 2025 cohort (1,319 patients annually). Under pooled EDF, the simulation produces case-loss fractions of $25.65\% \pm 1.90\%$ for early engagers and $28.68\% \pm 2.43\%$ for late engagers, with an aggregate case-loss fraction of $26.70\% \pm 1.74\%$. This corresponds to approximately 352.2 ± 31.2 case losses among $1,316.9 \pm 34.6$ patients; the CUH 2025 record of 362 case losses among 1,319 patients (27.4%) again falls within the simulated ± 1 SD band. The close agreement in both the in-sample and out-of-sample cohorts supports the use of the calibrated model for the prospective policy analysis that follows.

7.2. Main Results

In this section, we evaluate the calibrated CUH model. Section 7.2.1 compares pooled EDF with dedicated EDF under capacity reservation. Section 7.2.2 examines how the optimal reservation changes when advance surgery-date confirmation has patient-experience value. Section 7.2.3 shows how the same reservation policy enables early engagers to receive confirmed dates at clinical request.

7.2.1. Efficiency–Fairness Trade-off We first compare pooled EDF with dedicated EDF under capacity reservation, using $c^E = 1$, $c^L = 2$, and $v = 0$. For each equity tolerance $\Delta \in \{5\%, 10\%, 15\%, 20\%\}$, we compute the feasible cost-minimizing reservation fraction γ^* by grid search over $\gamma \in [0, 1]$ with step size 0.01. Because γ^*N need not be integer, we implement fractional reservations by stochastic rounding: in each week, N^E is rounded up with probability equal to its fractional part and rounded down otherwise, with $N^L = N - N^E$. This preserves the target reservation in expectation; in practice, a hospital could use fixed integer rounding, with rounding down N^E as the conservative choice when late-engager losses are more costly. Table 1 reports the simulated outcomes.

Table 1 Pooled EDF and dedicated EDF with capacity reservation at four equity tolerances Δ (2024 in-sample, 100 replications, $v = 0$, $c^E = 1$, $c^L = 2$).

Policy	γ^*	p^E (%)	p^L (%)	N^E	N^L	Pooled cost	Dedicated cost	Saving (%)
Pooled EDF	—	19.48 ± 2.14	22.17 ± 2.72	—	—	339.9 ± 44.5	—	—
Dedicated EDF ($\Delta = 20\%$)	0.58	29.03 ± 2.18	9.62 ± 2.98	10.44	7.56	339.9 ± 44.5	309.3 ± 40.1	$+8.9 \pm 4.0$
Dedicated EDF ($\Delta = 15\%$)	0.60	26.72 ± 2.22	12.69 ± 3.28	10.80	7.20	339.9 ± 44.5	317.4 ± 43.2	$+6.6 \pm 3.9$
Dedicated EDF ($\Delta = 10\%$)	0.62	24.43 ± 2.28	16.16 ± 3.39	11.16	6.84	339.9 ± 44.5	328.8 ± 44.2	$+3.2 \pm 4.0$
Dedicated EDF ($\Delta = 5\%$)	0.64	22.19 ± 2.35	19.93 ± 3.59	11.52	6.48	339.9 ± 44.5	343.2 ± 46.1	-1.1 ± 4.6

Note. Pooled cost and Dedicated cost report total case-loss cost C under pooled EDF and dedicated EDF, respectively. Saving (%) is the relative cost reduction of dedicated EDF over pooled EDF, computed as $(\text{Pooled cost} - \text{Dedicated cost}) / \text{Pooled cost}$.

The pooled EDF baseline appears only mildly imbalanced in case-loss fractions, with $p^E = 19.48\% \pm 2.14\%$ for early engagers and $p^L = 22.17\% \pm 2.72\%$ for late engagers (a 2.69pp gap). This near-parity is misleading. EDF equalizes urgency within a pooled queue, but it does not account for the higher cost of losing a late engager. As waiting early engagers approach their deadlines, they compete directly with newly arriving late engagers for the same slots, diluting the urgency protection that EDF is meant to provide.

Capacity reservation corrects this mismatch by shifting capacity toward late engagers relative to their demand share. At $\Delta = 20\%$, the optimal reservation is $\gamma^* = 0.58$, corresponding to 10.44 weekly slots for early engagers and 7.56 for late engagers. Because early engagers account for 64.9% of demand, this reservation deliberately protects late-engager capacity. The late-engager case-loss fraction falls from 22.17% to 9.62%, a 57% relative reduction, while the early-engager fraction rises from 19.48% to 29.03%. Total case-loss cost falls from 339.9 to 309.3 normalized units per year, an 8.9% reduction, or approximately £15,300 annually when one unit is valued at £500.

The result is not that reservation eliminates case losses; total capacity remains fixed. Rather, reservation reallocates scarce elective-caesarean capacity toward the patients for whom case loss is more costly. This is exactly the structure predicted by Theorem 2: the optimal split equalizes marginal capacity value across the two dedicated queues. Since $c^L = 2c^E$, the cost-minimizing policy reserves more capacity for late engagers than their volume share alone would imply.

Tightening the equity tolerance changes the nature of the trade-off. As Δ decreases from 20% to 5%, the optimal reservation shifts from $\gamma^* = 0.58$ to 0.64, pulling the early and late case-loss fractions closer together. At $\Delta = 5\%$, the two fractions are nearly equal, with $p^E = 22.19\%$ and $p^L = 19.93\%$. This greater parity comes at a cost: the cost under dedicated EDF rises to 343.2 units, making it slightly more expensive than pooled EDF. Thus, stricter equality in case-loss fractions can conflict with clinical efficiency when case losses are differentially costly. Section 7.3.1 examines how this trade-off changes as the cost ratio varies.

7.2.2. Effect of Patient Satisfaction The preceding analysis treats the planner’s objective as purely loss-driven. In practice, however, early engagers also incur a patient-experience cost from waiting roughly ten weeks between clinical request and surgery-date confirmation. The extended objective $\tilde{C}(\gamma) = C(\gamma) - \nu\gamma N$ credits the satisfaction value of reserved early-engager capacity, where ν is the normalized value of confirming an early engager’s surgery date at clinical request. We solve the extended problem for $\nu \in \{0, 0.1, 0.3, 0.5, 0.7, 1.0\}$ at $\Delta = 20\%$. Table 2 reports the results.

Table 2 Effect of satisfaction value ν ($\Delta = 20\%$, $c^E = 1$, $c^L = 2$, 2024 in-sample, 100 replications).

ν	$\tilde{\gamma}^*$	p^E (%)	p^L (%)	N^E	N^L	Pooled ext. cost	Dedicated ext. cost	Saving (%)
0.0	0.58	29.03 $\pm$ 2.18	9.62 $\pm$ 2.98	10.44	7.56	339.9 $\pm$ 44.5	309.3 $\pm$ 40.1	+8.9 $\pm$ 4.0
0.1	0.58	29.03 $\pm$ 2.18	9.62 $\pm$ 2.98	10.44	7.56	339.9 $\pm$ 44.5	255.0 $\pm$ 40.1	+25.1 $\pm$ 4.0
0.3	0.58	29.03 $\pm$ 2.18	9.62 $\pm$ 2.98	10.44	7.56	339.9 $\pm$ 44.5	146.5 $\pm$ 40.1	+57.6 $\pm$ 6.7
0.5	0.60	26.72 $\pm$ 2.22	12.69 $\pm$ 3.28	10.80	7.20	339.9 $\pm$ 44.5	36.6 $\pm$ 43.2	+90.6 $\pm$ 11.4
0.7	0.61	25.56 $\pm$ 2.21	14.35 $\pm$ 3.36	10.98	7.02	339.9 $\pm$ 44.5	-77.4 $\pm$ 44.0	+124.8 $\pm$ 16.0
1.0	0.70	15.33 $\pm$ 2.44	32.48 $\pm$ 3.38	12.60	5.40	339.9 $\pm$ 44.5	-259.7 $\pm$ 47.8	+179.4 $\pm$ 24.1

Note. For $\nu > 0$, savings should be interpreted as extended-objective improvements, not pure case-loss cost reductions.

A positive satisfaction value shifts the optimal reservation toward early engagers, as predicted by the first-order condition (11). The shift is modest for small ν : $\tilde{\gamma}^*$ remains 0.58 for $\nu \in \{0, 0.1, 0.3\}$ and rises to 0.60 at $\nu = 0.5$. It becomes substantial only when the satisfaction value is large, reaching 0.70 at $\nu = 1.0$. The flat regions reflect the grid search; with a continuous search, the optimum moves smoothly with ν .

The welfare effect is much larger than the movement in $\tilde{\gamma}^*$ suggests. Even at $\nu = 0.1$, equal to only one-tenth of the early-engager case-loss cost, the extended-objective improvement rises from 8.9% to 25.1%. At $\nu = 0.5$, it reaches 90.6%; at $\nu = 1.0$, it reaches 179.4%. These large gains arise because satisfaction value is unlocked by capacity that is already reserved for operational reasons. In other words, reservation provides a dual benefit: it reduces costly late-engager case losses and converts early-engager advance information into earlier date confirmation.

7.2.3. Advance Confirmation The preceding results assume the current confirmation rule in which b_i is set near the start of the allocated surgery week. Corollary 1 shows that this timing choice is not operationally necessary. For any served patient, moving the confirmation date earlier within the interval $[r_i, s_i]$ leaves the case-loss fractions, cost, and optimal reservation unchanged. The planner can therefore set $b_i = r_i$ for early engagers served through the reserved capacity, confirming their surgery dates at clinical request without changing capacity reservation or case-loss outcomes. Since late engagers request close to surgery and are confirmed soon after request, the experiential gain is concentrated among early engagers.

At the baseline reservation $\gamma^* = 0.58$, dedicated EDF reserves $N^E = 10.44$ weekly slots for early engagers. This capacity serves approximately 70% of the $\lambda^E = 15.0$ early-engager requests per week, or about 543 patients annually. Under advance confirmation, every early engager served through the reserved capacity receives a confirmed surgery date at clinical request, compared with none under current practice.

The total time from request to surgery is unchanged. For early engagers served under either policy, this interval averages 77.8 days. What changes is its informational content. Under current practice, an early engager waits about 74.3 days for confirmation and then has only 3.5 days of confirmed notice. Under advance confirmation, an early engager served through the reserved capacity has zero pre-confirmation wait, and the full 77.8 days become confirmed notice. Thus, the policy does not require more theatre capacity or earlier surgery dates; it simply converts latent advance information into a confirmed delivery plan.

7.3. Sensitivity Analyses

We test whether the dedicated-EDF policy is a finely tuned artifact of the baseline calibration. It is not. Across five perturbations—case-loss cost ratio, early/late classification cutoff, seasonal demand, no-shows, and emergency spillover—the optimal reservation remains stable or shifts in economically predictable directions. Unless stated otherwise, we use the baseline parameters $\Delta = 20\%$, $v = 0$, $c^E = 1$, $c^L = 2$, and $N = 18$; the swept ranges are summarized in Appendix EC.4.1.

7.3.1. Case-Loss Cost Ratio We first vary the cost ratio c^L/c^E over the calibrated range $[1.31, 2.49]$, holding $c^E + c^L = 3$, and include $c^L/c^E = 3.0$ as an upper robustness point. The range is derived from NHS Reference Costs and clinical-pathway differentials, including length of stay, anaesthesia mix, and NICU admission, as detailed in Appendix EC.4.1.2. Table 3 reports the results.

c^L/c^E	γ^*	p^E (%)	p^L (%)	N^E	N^L	Pooled cost	Dedicated cost	Saving (%)
1.31	0.60	26.72 ± 2.22	12.69 ± 3.28	10.80	7.20	357.5 ± 45.9	363.8 ± 44.4	−1.9 ± 3.5
2.00	0.58	29.03 ± 2.18	9.62 ± 2.98	10.44	7.56	339.9 ± 44.5	309.3 ± 40.1	+8.9 ± 4.0
2.49	0.58	29.03 ± 2.18	9.62 ± 2.98	10.44	7.56	331.7 ± 43.9	283.3 ± 39.7	+14.5 ± 4.3
3.00	0.58	29.03 ± 2.18	9.62 ± 2.98	10.44	7.56	325.2 ± 43.6	262.9 ± 39.5	+19.2 ± 4.7

The reservation fraction is nearly unchanged across the sweep, remaining between 0.58 and 0.60. What changes is the value of protection. When late-engager losses are only modestly more costly than early-engager

losses, reservation offers little cost advantage. As the cost ratio rises, the same capacity split becomes increasingly valuable: savings move from -1.9% at $c^L/c^E = 1.31$ to $+19.2\%$ at $c^L/c^E = 3.0$. Thus, the policy's economic value grows precisely when clinical differentiation matters most.

7.3.2. Early/Late Engager Classification Cutoff The early/late cutoff determines which information lead times $I_i = e_i - r_i$ are classified as early. The baseline cutoff is 35 days, which yields an early-engager share of 64.9%. We vary the cutoff over $\{20, 25, 30, 35, 40\}$ days, generating early shares of $\{80.6\%, 75.9\%, 71.2\%, 64.9\%, 59.2\%\}$. The cutoff shifts the composition of early-engager versus late-engager demand: a lower (higher) cutoff classifies more patients as early (late). Table 4 reports the results.

Table 4 Sensitivity to the early/late engager classification cutoff (in days), holding $\lambda = 23.11$ patients/week and the indication mix fixed ($\Delta = 20\%$, $v = 0$, $c^E = 1$, $c^L = 2$, 2024 in-sample, 100 replications).

Cutoff (days)	γ^*	p^E (%)	p^L (%)	N^E	N^L	Pooled cost	Dedicated cost	Saving (%)
20	0.75	26.03 ± 2.09	9.21 ± 3.19	13.50	4.50	298.3 ± 38.6	296.5 ± 33.9	$+0.3 \pm 4.6$
25	0.69	27.67 ± 2.07	7.92 ± 3.00	12.42	5.58	311.1 ± 40.6	299.7 ± 34.7	$+3.3 \pm 5.0$
30	0.65	27.49 ± 2.08	9.78 ± 3.12	11.70	6.30	323.6 ± 42.4	304.5 ± 38.3	$+5.7 \pm 4.1$
35	0.58	29.03 ± 2.18	9.62 ± 2.98	10.44	7.56	339.9 ± 44.5	309.3 ± 40.1	$+8.9 \pm 4.0$
40	0.52	30.19 ± 2.29	10.17 ± 3.12	9.36	8.64	354.4 ± 46.4	316.4 ± 43.7	$+10.7 \pm 3.7$

The optimal reservation tracks the composition of the patient stream. As the cutoff becomes more permissive (smaller cutoff) and the early-engager share rises, γ^* increases from 0.52 at a 40-day cutoff to 0.75 at a 20-day cutoff. The cost saving moves in the opposite direction, rising from essentially zero at the most permissive cutoff to 10.7% at the strictest 40-day cutoff. The reason is simple: when many marginal patients are classified as early, the late-engager pool is small and there is less value in protecting it. As the cutoff becomes stricter, the late-engager pool grows and the value of dedicated late-engager capacity increases. At the institutional 35-day threshold, the dedicated EDF policy delivers a clear cost benefit.

7.3.3. Time-Varying Demand We next test seasonal demand variation using a sinusoidal weekly arrival rate, $\lambda_t = \lambda(1 + a \sin(2\pi t/52))$, with baseline mean $\lambda = 23.11$ and amplitude $a \in \{0.05, 0.10, 0.15, 0.20, 0.25\}$. CUH 2024 monthly delivery counts exhibit a peak-to-trough swing of approximately $\pm 17\%$, so this range covers the observed variation. Table 5 reports the results.

Table 5 Sensitivity to seasonal demand $\lambda_t = \lambda(1 + a \sin(2\pi t/52))$ ($\Delta = 20\%$, $v = 0$, $c^E = 1$, $c^L = 2$, 2024 in-sample, 100 replications).

Amplitude a	γ^*	p^E (%)	p^L (%)	N^E	N^L	Pooled cost	Dedicated cost	Saving (%)
0.05	0.58	28.33 ± 2.29	9.59 ± 2.76	10.44	7.56	329.4 ± 42.8	302.3 ± 35.5	$+8.0 \pm 4.7$
0.10	0.58	28.11 ± 2.30	10.04 ± 2.77	10.44	7.56	329.2 ± 41.0	304.9 ± 34.9	$+7.1 \pm 4.7$
0.15	0.57	28.97 ± 2.43	9.20 ± 2.79	10.26	7.74	329.8 ± 45.5	305.4 ± 37.3	$+7.0 \pm 5.6$
0.20	0.57	28.86 ± 2.42	10.07 ± 2.91	10.26	7.74	332.0 ± 46.1	312.0 ± 39.2	$+5.7 \pm 5.3$
0.25	0.56	29.77 ± 2.27	9.86 ± 2.63	10.08	7.92	334.4 ± 42.2	316.7 ± 35.6	$+5.0 \pm 5.3$

Seasonality has little effect on the recommended reservation. Across the full amplitude range, the cost-minimizing reservation γ^* stays between 0.56 and 0.58. Relative savings decline mildly, from 8.0% at $a = 0.05$ to 5.0% at $a = 0.25$, because seasonal peaks create short periods of overload that a fixed annual reservation cannot fully smooth. Even at the largest amplitude tested, however, dedicated EDF retains a positive cost advantage. The policy therefore does not require seasonal recalibration to remain effective.

7.3.4. No-Shows We introduce a per-patient no-show probability $p_{\text{ns}} \in \{0, 0.02, 0.05\}$. A no-show wastes the allocated slot and is counted as a case loss. This range is conservative for elective caesareans: outpatient no-show rates can be much higher (Dantas et al. 2018), but scheduled obstetric surgery involves stronger patient commitment, clinical preparation, and family logistics. Table 6 reports the results.

Table 6 Sensitivity to the per-patient no-show probability p_{ns} ($\Delta = 20\%$, $v = 0$, $c^E = 1$, $c^L = 2$, 2024 in-sample, 100 replications).

p_{ns}	γ^*	p^E (%)	p^L (%)	N^E	N^L	Pooled cost	Dedicated cost	Saving (%)
0.00	0.58	29.03 ± 2.18	9.62 ± 2.98	10.44	7.56	339.9 ± 44.5	309.3 ± 40.1	$+8.9 \pm 4.0$
0.02	0.58	31.09 ± 2.21	11.64 ± 3.06	10.44	7.56	373.4 ± 46.1	342.4 ± 41.0	$+8.1 \pm 4.5$
0.05	0.58	34.19 ± 2.30	14.60 ± 3.08	10.44	7.56	424.7 ± 49.0	391.6 ± 42.2	$+7.6 \pm 4.4$

No-shows increase case losses under both policies, but they do not change the reservation logic. As p_{ns} rises from 0 to 0.05, pooled cost increases from 339.9 to 424.7 normalized units, while dedicated cost increases from 309.3 to 391.6. The reservation fraction remains fixed at $\gamma^* = 0.58$, and the relative saving narrows only modestly, from 8.9% at $p_{\text{ns}} = 0$ to 7.6% at $p_{\text{ns}} = 0.05$. Because no-shows remove effective capacity from both systems, they raise the level of loss but preserve the advantage of capacity reservation.

7.3.5. Emergency Case Spillover Finally, we model emergency spillover as a deterministic reduction in effective elective capacity, with $N \in \{16, 17, 18\}$. This captures weeks in which emergency caesareans consume capacity otherwise available to the elective list. The reservation fraction is re-optimized at each capacity level. Table 7 reports the results.

Table 7 Sensitivity to emergency case spillover, modeled as a deterministic reduction in the weekly elective capacity N ($\Delta = 20\%$, $v = 0$, $c^E = 1$, $c^L = 2$, 2024 in-sample, 100 replications).

N	γ^*	p^E (%)	p^L (%)	N^E	N^L	Pooled cost	Dedicated cost	Saving (%)
18	0.58	29.03 ± 2.18	9.62 ± 2.98	10.44	7.56	339.9 ± 44.5	309.3 ± 40.1	$+8.9 \pm 4.0$
17	0.58	32.76 ± 2.09	13.24 ± 3.29	9.86	7.14	406.3 ± 45.2	369.1 ± 43.6	$+9.2 \pm 3.2$
16	0.58	36.53 ± 2.00	17.52 ± 3.46	9.28	6.72	473.6 ± 45.5	434.6 ± 45.4	$+8.2 \pm 2.9$

Emergency spillover worsens overload, but it does not overturn the reservation structure. As weekly elective capacity falls from 18 to 16 slots, both pooled and dedicated costs rise sharply. Yet γ^* remains 0.58, the absolute saving stays between 30 and 39 normalized units per year, and the relative saving remains

between 8.2% and 9.2%. The implication is operationally important: when emergency demand encroaches on elective capacity, the need for reservation becomes more pressing, not less. The same capacity split continues to protect late engagers without requiring retuning.

7.4. Managerial Recommendations

The managerial implication is simple: CUH should not rely on a single pooled EDF list to manage elective-caesarean demand. Pooled EDF creates two operational failures. It does not adequately protect late engagers, whose case losses are clinically more costly, and it does not convert early engagers' advance information into advance certainty. The recommended policy is therefore to partition the existing weekly theatre list into dedicated EDF queues, reserving approximately 10 of 18 weekly slots for early engagers and 8 for late engagers, and to confirm surgery dates at request for early engagers served through the reserved capacity.

This policy uses the same theatre capacity but changes its allocation. Its main effects are as follows.

- *Late-engager case losses fall by more than half.* The late-engager case-loss fraction drops from 22.17% under pooled EDF to 9.62% under dedicated EDF, a 57% relative reduction. In the 2024 cohort of approximately 422 late engagers, this corresponds to roughly 53 avoided late-engager case losses per year. These are precisely the cases for which missed planned capacity is most likely to trigger higher-risk emergency conversion.
- *Total case-loss cost falls without adding capacity.* At the baseline cost ratio $c^L = 2c^E$, total normalized case-loss cost falls from 339.9 to 309.3 units per year, an 8.9% reduction, or approximately £15,300 annually. The saving increases to 14.5%–19.2% when late-engager case losses are valued at the upper end of the calibrated cost range. The policy does raise the early-engager case-loss fraction, from 19.48% to 29.03%, corresponding to about 1.4 additional early-engager case losses per week (74 annually) on the 780 early-engager cohort per year. This trade-off is acceptable on clinical grounds because each early-engager case loss carries approximately half the clinical cost of a late-engager case loss.
- *Early engagers gain meaningful advance certainty.* The reserved early-engager capacity serves approximately 70% of early-engager requests, or about 543 patients per year. These patients can receive a confirmed surgery date at clinical request rather than waiting until the start of the allocated surgery week. As a consequence, average confirmed notice increases from approximately 3.5 days under current practice to about 77.8 days under advance confirmation. Even a modest satisfaction value, $v = 0.1$, raises the policy's annual value from approximately £15,300 to £42,450.

Taken together, the sensitivity analyses and 2025 held-out cohort show that the recommendation is not finely tuned to the baseline calibration. Except when the early/late cutoff changes the patient mix by design, the optimal reservation remains close to $\gamma^* = 0.58$. The 2025 cohort yields a nearly identical recommendation, $\gamma^* = 0.59$, with an annual saving of approximately 40.0 normalized units, or £20,000. Savings increase when late-engager losses are more costly or the late-engager pool is more sharply identified, and remain positive under seasonal demand, no-shows, and emergency spillover.

Implementation is operationally modest. CUH need not add theatre capacity, replace EDF, or adopt a complex dynamic scheduling algorithm. The change is to make the capacity split explicit: maintain a weekly early-engager list of approximately 10 slots and a late-engager list of approximately 8 slots, apply EDF within each list, and confirm surgery dates at clinical request for early engagers served through the reserved capacity. This turns an implicit pooled queue into an auditable reservation rule that protects time-critical late demand while giving many early engagers the certainty that their advance request should make possible.

At NHS scale, the same proportional reduction would be meaningful. Applying the CUH saving to approximately 150,000 elective caesareans annually—roughly 60% of the 244,005 caesarean births, with the remainder performed as emergencies—implies savings of about £1.9 million per year, alongside fewer high-risk emergency conversions and improved patient experience. The managerial lesson is: under overload, elective-caesarean capacity should be partitioned explicitly rather than pooled implicitly.

8. Conclusion

This paper studies elective caesarean scheduling in the English NHS, where demand for theatre slots exceeds capacity and some patients are diverted from the elective pathway to emergency delivery. We establish three results. First, in an overloaded pooled system with two-sided clinical windows, steady-state case losses for both patient types collapse to a common service-time threshold determined only by workload mix and capacity. EDF prioritizes imminent deadlines, but pooling causes early engagers to age into the same urgency range as late engagers, while giving early engagers little benefit from advance request. The resulting inequity is structural: within-queue scheduling can redistribute case losses across patient types, but it cannot remove the underlying capacity conflict. Second, we propose a capacity-reservation policy that partitions theatre slots between early and late engagers and derive the optimal reservation fraction γ^* in closed form. The optimal split equalizes the marginal value of reserved capacity across the two dedicated queues. Because the reservation depends on neither clinical-window nor information-lead-time distributions, every early engager assigned a reserved slot can receive a confirmed surgery date at the time of clinical request without changing capacity reservation or case-loss outcomes. Moreover, any hospital that assigns positive value to advance confirmation should reserve more capacity for early engagers. Third, calibrated to Cambridge University Hospitals data (1,202 elective caesareans, 18 weekly slots, and a demand-to-supply ratio of 1.32), the policy reserves $\gamma^*N = 10.44$ slots for early engagers and 7.56 for late engagers. This reduces late-engager case losses from 22.17% to 9.62%, a 57% relative reduction, while providing immediate surgery-date confirmation to approximately 543 early engagers per year, compared with none under current practice. Total case-loss cost falls by 8.9%, or approximately £15,300 annually in the Cambridge calibration. Applied to approximately 150,000 elective caesareans performed annually in the English NHS, the same proportional saving would amount to roughly £1.9 million per year.

The managerial recommendation is concrete. NHS maternity units should replace a single pooled elective-caesarean list with two auditable EDF lists: approximately 10 weekly slots for early engagers and 8 for

late engagers in the Cambridge setting. Early engagers served through the reserved capacity should receive confirmed surgery dates at clinical request. The policy requires no additional theatre capacity, no new staffing, and no complex dynamic scheduling algorithm. It simply makes the capacity split explicit, using information already available to scheduling teams. This design operationalizes the RCOG Good Practice Paper’s call for dedicated planned-caesarean operating sessions sized to demand, and gives RCOG and NICE policymakers a closed-form rule that any unit can implement using its own demand and capacity data.

Two follow-on studies are natural. The first is a prospective field experiment at Cambridge University Hospitals, which is currently being prepared. Such a study would allow the reservation rule to be evaluated in practice, including its effects on case losses, confirmation timing, and patient experience. It would also provide data to calibrate the patient-experience value v and refine the per-type case-loss cost estimates. The second is multi-site validation across NHS maternity units. Because local case mix, demand pressure, and theatre capacity vary across hospitals, a multi-site study could examine how γ^* changes across operating environments and assess whether our proposed reservation policy can support broader implementation.

Endnotes

¹The formal definition of “well-defined” in this context and the full mathematical statement of Assumption 3 are provided in Appendix EC.1.1.2.

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Electronic Companion

In this Electronic Companion, we provide supporting technical material for the paper “Reserving Capacity, Restoring Fairness: Scheduling Elective Caesarean Sections in the English NHS”.

EC.1. Proofs from Section 4

EC.1.1. Preliminaries

This subsection collects notation for the pre-limit stochastic system, formal statements of the two fluid-scaling assumptions, and two auxiliary lemmas that are invoked in multiple proofs below.

EC.1.1.1. Pre-limit notation In the n -th system described in Section 4.1, the system state is $\Phi_t^{(n)} := (\Phi_t^{Q(n)}, \Phi_t^{S(n)})$, where $\Phi_t^{Q(n)}$ and $\Phi_t^{S(n)}$ are counting measures on $\mathcal{R}_0$ recording the residual pairs (σ, τ) of waiting and in-service requests, respectively, with $\Phi_t^{S(n)}(\mathcal{R}_0) \leq N^{(n)}$.

Since EDF prioritizes by remaining time-to-deadline τ , we define the τ -marginal of the waiting measure by $\Phi_t^{Q(n), \tau}(I) := \Phi_t^{Q(n)}(\mathbb{R}_+ \times I)$ for any Borel set $I \subseteq \mathbb{R}_+$. The *head-of-line (HOL) time-to-deadline* is

$$c^{(n)}(t) := \begin{cases} \sup \{x > 0 : \Phi_t^{Q(n), \tau}([0, x)) = 0\}, & \text{if } Q_t^{(n)} > 0, \\ 0, & \text{if } Q_t^{(n)} = 0, \end{cases}$$

where $Q_t^{(n)} := \Phi_t^{Q(n)}(\mathcal{R}_0)$. Under EDF, $c^{(n)}(t)$ is the smallest remaining time-to-deadline among all waiting requests.

The *frontier* process, following Atar et al. (2015), is

$$F^{(n)}(t) := \max \left\{ \sup_{s \leq t, Q_s^{(n)} > 0} \{c^{(n)}(s) - (t - s)\}, F^{(n)}(0) - t \right\},$$

and the *current time-to-deadline* is $C^{(n)}(t) := c^{(n)}(t)$ when $Q_t^{(n)} > 0$ and $C^{(n)}(t) := F^{(n)}(t)$ when $Q_t^{(n)} = 0$.

The frontier acts as a high-water mark: any request that entered service did so with a remaining time-to-deadline at or below the frontier at that moment, so $F^{(n)}(t)$ provides an upper bound on the remaining times-to-deadline of all in-service requests.

For any τ -interval $I \in \mathcal{B}((0, \infty))$, the *queue workload* in I is

$$W_t^{(n)}(I) := \int_{(0, \infty) \times I} \sigma \Phi_t^{Q(n)}(d\sigma, d\tau).$$

EC.1.1.2. Formal statement of Assumption 3 Let $I_0^{(n)}$ index the jobs initially present in the queue of the n -th system at time 0; each such job i carries a pair $(S_i^{(n)}, T_i^{(n)}) \in \mathcal{R}_0$ recording its service requirement and remaining time-to-deadline at $t = 0$. Let $\mathcal{M}_f(\mathcal{R}_0)$ be the space of finite Borel measures on $\mathcal{R}_0$, equipped with the topology of weak convergence. Assumption 3 requires the following conditions:

- (1) **Initial queue and frontier.** There exists a pair $(q_0, F(0)) \in \mathcal{M}_f(\mathcal{R}_0) \times [0, \bar{\tau}]$ such that, for the τ -marginal q_0^τ , we have $q_0^\tau([0, F(0)]) = 0$ and $q_0^\tau([0, x]) > 0$ for all $x > F(0)$. Moreover,

$$\left(\frac{1}{n} q_0^{(n)}, F^{(n)}(0) \right) \xrightarrow{d} (q_0, F(0)) \quad \text{as } n \rightarrow \infty.$$

- (2) **Initial LLN.** There exists a finite constant $\kappa_0 > 0$ such that $|I_0^{(n)}|/n \rightarrow \kappa_0$ as $n \rightarrow \infty$, and for every bounded measurable $\psi : \mathbb{R}_+^2 \rightarrow \mathbb{R}$,

$$\frac{1}{|I_0^{(n)}|} \sum_{i \in I_0^{(n)}} \psi(S_i^{(n)}, T_i^{(n)}) \xrightarrow{P} \mathbb{E}[\psi(S, T)] \quad \text{as } n \rightarrow \infty.$$

EC.1.1.3. Formal statement of Assumption 4 Let $\mathcal{M}_f(\mathcal{R}_0)$ be as above. Assumption 4 requires that there exist deterministic càdlàg functions $t \mapsto q_t \in \mathcal{M}_f(\mathcal{R}_0)$, $t \mapsto s_t \in \mathcal{M}_f(\mathcal{R}_0)$, and $t \mapsto F(t) \in \mathbb{R}$, such that for every finite horizon $T_0 > 0$, as $n \rightarrow \infty$, the triple

$$\left(q^{(n)}(\cdot), s^{(n)}(\cdot), F^{(n)}(\cdot) \right) \Big|_{[0, T_0]}$$

converges in distribution to $(q(\cdot), s(\cdot), F(\cdot))$ in the Skorokhod J_1 topology on $D([0, T_0], \mathcal{M}_f(\mathcal{R}_0) \times \mathcal{M}_f(\mathcal{R}_0) \times \mathbb{R})$, where the state space carries the product topology induced by weak convergence on $\mathcal{M}_f(\mathcal{R}_0)$. The limiting dynamics $(q(\cdot), s(\cdot))$ are governed by the non-preemptive EDF rule.

Under this assumption, the τ -marginal processes also converge on every finite horizon: $q^{(n), \tau}(\cdot) \Rightarrow q^\tau(\cdot)$ and $s^{(n), \tau}(\cdot) \Rightarrow s^\tau(\cdot)$ as $n \rightarrow \infty$. This follows from the continuous mapping theorem, since $\mu \mapsto \mu^\tau$ is continuous under weak convergence.

EC.1.1.4. Reusable auxiliary results The following two lemmas are invoked in multiple proofs within the vanishing-gap argument.

LEMMA EC.1 (Arrivals LLN). Fix $T_0 > 0$. For every bounded measurable $\psi : \mathbb{R}_+^2 \rightarrow \mathbb{R}$,

$$\sup_{t \in [0, T_0]} \left| \frac{1}{n} \sum_{i: a_i^{(n)} \leq t} \psi(S_i^{(n)}, T_i^{(n)}) - \lambda t \mathbb{E}[\psi(S, T)] \right| \xrightarrow{P} 0 \quad \text{as } n \rightarrow \infty.$$

Proof. Fix a bounded measurable ψ with $\|\psi\|_\infty \leq M < \infty$, and define $X^{(n)}(t) := \sum_{i: a_i^{(n)} \leq t} \psi(S_i^{(n)}, T_i^{(n)})$. Because arrivals are Poisson with rate $n\lambda$ and marks are i.i.d. independent of arrival times, $\{X^{(n)}(t)\}_{t \geq 0}$ is a compound Poisson process with rate $n\lambda$ and jump-size distribution $\psi(S, T)$. The centered process $\tilde{X}^{(n)}(t) := X^{(n)}(t) - n\lambda t \mathbb{E}[\psi(S, T)]$ is a càdlàg martingale with $\mathbb{E}[(\tilde{X}^{(n)}(T_0))^2] = n\lambda T_0 \mathbb{E}[(\psi(S, T))^2] \leq n\lambda T_0 M^2$. By Doob's maximal inequality, for every $\epsilon > 0$,

$$\Pr \left(\sup_{t \in [0, T_0]} |\tilde{X}^{(n)}(t)| \geq n\epsilon \right) \leq \frac{\lambda T_0 M^2}{n\epsilon^2} \rightarrow 0.$$

Dividing by n gives the result. $\square$

REMARK EC.1. By Lemma EC.1, for any bounded measurable ψ there exists a sequence $\varepsilon_A(n) \rightarrow 0$ such that

$$\Pr\left(\sup_{t \in [0, T_0]} \left| \frac{1}{n} \sum_{i: a_i^{(n)} \leq t} \psi(S_i^{(n)}, T_i^{(n)}) - \lambda t \mathbb{E}[\psi(S, T)] \right| \leq \varepsilon_A(n)\right) \geq 1 - \varepsilon_A(n).$$

LEMMA EC.2 (**Processed service-capacity bound**). Let $U^{(n)}(t) := \int_0^t B^{(n)}(s) ds$ denote the cumulative processed work by time t in the n -th system, where $B^{(n)}(s) \in [0, nN]$ denotes the number of busy servers at time s . Then $U^{(n)}(t) \leq nNt$ for all $t \geq 0$. Moreover, if the system is busy (all servers occupied) on $[0, T_0]$, then $n^{-1}U^{(n)}(t) = Nt$ for every $t \in [0, T_0]$.

Proof. The bound $U^{(n)}(t) = \int_0^t B^{(n)}(s) ds \leq nNt$ follows immediately from $B^{(n)}(s) \leq nN$. When the system is busy on $[0, T_0]$, $B^{(n)}(s) = nN$ for all $s \in [0, T_0]$, so $n^{-1}U^{(n)}(t) = Nt$. $\square$

EC.1.2. Proof of Lemma 1 (Threshold separation)

The proof requires the following intermediate result, which establishes that the stochastic gap between the head-of-line time-to-deadline and the frontier vanishes at the fluid scale.

LEMMA EC.3 (**Vanishing gap**). Fix $T_0 > 0$. For any $\eta > 0$,

$$\lim_{n \rightarrow \infty} \Pr\left(\sup_{t \in [0, T_0]} q_t^{(n), \tau}\left([C^{(n)}(t), F^{(n)}(t)]\right) > \eta\right) = 0.$$

Assuming Lemma EC.3, we prove Lemma 1.

Proof of Lemma 1. Fix $t > 0$. Define $\tau_t := F(t)$, where $F(\cdot)$ is the deterministic fluid limit of $F^{(n)}(\cdot)$ under Assumption 4. We record two properties that hold for every n and t in the pre-limit system:

- (i) *No waiting mass below $C^{(n)}(t)$.* If the queue is nonempty, then $C^{(n)}(t) = c^{(n)}(t)$ is the smallest remaining time-to-deadline in the queue, so

$$\Phi_t^{Q^{(n), \tau}\left([0, C^{(n)}(t)]\right)} = 0. \quad (\text{EC.1})$$

- (ii) *No in-service mass above $F^{(n)}(t)$.* Any job in service at time t entered service at some time $u \leq t$ with HOL time-to-deadline $c^{(n)}(u-)$. Its remaining time-to-deadline at time t is $c^{(n)}(u-) - (t - u) \leq F^{(n)}(t)$ by definition of the frontier. Therefore,

$$\Phi_t^{S^{(n), \tau}\left((F^{(n)}(t), \infty)\right)} = 0. \quad (\text{EC.2})$$

By Lemma EC.3, the fluid-scaled waiting mass in the gap vanishes:

$$q_t^{(n), \tau}\left([C^{(n)}(t), F^{(n)}(t)]\right) \rightarrow 0 \quad \text{in probability as } n \rightarrow \infty. \quad (\text{EC.3})$$

Combining (EC.1) and (EC.3) gives

$$q_t^{(n), \tau}\left([0, F^{(n)}(t)]\right) = q_t^{(n), \tau}\left([0, C^{(n)}(t)]\right) + q_t^{(n), \tau}\left([C^{(n)}(t), F^{(n)}(t)]\right) \rightarrow 0 \quad (\text{EC.4})$$

in probability. Under Assumption 4, for each $t > 0$,

$$q_t^{(n),\tau} \Rightarrow q_t^\tau, \quad s_t^{(n),\tau} \Rightarrow s_t^\tau, \quad F^{(n)}(t) \rightarrow \tau_t. \quad (\text{EC.5})$$

Step 1: $q_t^\tau([0, \tau_t]) = 0$. Fix $\epsilon > 0$. By (EC.5), $\Pr(F^{(n)}(t) \geq \tau_t - \epsilon) \rightarrow 1$. On this event, $[0, \tau_t - \epsilon] \subset [0, F^{(n)}(t)]$, so $q_t^{(n),\tau}([0, \tau_t - \epsilon]) \leq q_t^{(n),\tau}([0, F^{(n)}(t)])$. By (EC.4), the right-hand side converges to 0, hence $q_t^{(n),\tau}([0, \tau_t - \epsilon]) \rightarrow 0$ in probability.

Since q_t^τ is a finite measure, it has at most countably many atoms. For all but countably many $\epsilon > 0$, the set $[0, \tau_t - \epsilon]$ is a q_t^τ -continuity set. By (EC.5) and the Portmanteau theorem for continuity sets, $q_t^{(n),\tau}([0, \tau_t - \epsilon]) \rightarrow q_t^\tau([0, \tau_t - \epsilon])$. Since the left-hand side also converges to 0, we conclude $q_t^\tau([0, \tau_t - \epsilon]) = 0$ for a dense set of $\epsilon > 0$. Monotonicity extends this to all $\epsilon > 0$, and letting $\epsilon \downarrow 0$ gives $q_t^\tau([0, \tau_t]) = 0$.

Step 2: $s_t^\tau((\tau_t, \infty)) = 0$. Fix $\epsilon > 0$. By (EC.5), $\Pr(F^{(n)}(t) \leq \tau_t + \epsilon) \rightarrow 1$. On this event, $(\tau_t + \epsilon, \infty) \subset (F^{(n)}(t), \infty)$, and (EC.2) gives $s_t^{(n),\tau}((\tau_t + \epsilon, \infty)) = 0$ with probability tending to 1. By (EC.5) and the Portmanteau theorem for the open set $(\tau_t + \epsilon, \infty)$, $s_t^\tau((\tau_t + \epsilon, \infty)) \leq \liminf_{n \rightarrow \infty} s_t^{(n),\tau}((\tau_t + \epsilon, \infty)) = 0$. Writing $(\tau_t, \infty) = \bigcup_{m \geq 1} (\tau_t + 1/m, \infty)$ and using monotonicity of measures gives $s_t^\tau((\tau_t, \infty)) = 0$. $\square$

To complete the proof of Lemma 1, we establish the vanishing-gap property (Lemma EC.3). The proof requires the following auxiliary results, whose proofs are deferred to the end of this subsection.

LEMMA EC.4 (Auxiliary threshold). *There exists a unique $\bar{\tau} \in (T_{\min}, T_{\max})$ satisfying $N = \lambda \mathbb{E}[S\mathbf{1}_{\{T \leq \bar{\tau}\}}]$.*

LEMMA EC.5 (Initial workload convergence). *There exists $W_0 \in \mathcal{M}_f(\mathbb{R}_+)$ with $W_0([0, F(0)]) = 0$ and $W_0([0, x]) > 0$ for all $x > F(0)$, such that $n^{-1}W_0^{(n)} \Rightarrow W_0$ as $n \rightarrow \infty$.*

LEMMA EC.6 (Negligibility of boundary-slack workload). *Define*

$$\tilde{G}_0^{(n)}(\xi) := \sum_{i \in I_0^{(n)}} S_i^{(n)} \mathbf{1}_{\{0 \leq T_i^{(n)} - S_i^{(n)} < \xi\}}.$$

For any $x > 0$,

$$\lim_{\xi \downarrow 0} \limsup_{n \rightarrow \infty} \Pr\left(\frac{1}{n} \tilde{G}_0^{(n)}(\xi) > x\right) = 0.$$

LEMMA EC.7 (Frontier upper bound). *Let $\bar{\tau}$ be the unique solution to $N = \lambda \mathbb{E}[S\mathbf{1}_{\{T \leq \bar{\tau}\}}]$ (existence guaranteed by Lemma EC.4). For any $T_0, \epsilon > 0$,*

$$\lim_{n \rightarrow \infty} \Pr\left(\sup_{t \in [0, T_0]} F^{(n)}(t) > \bar{\tau} + \epsilon\right) = 0.$$

LEMMA EC.8 (Vanishing workload in the gap). *Fix $T_0 > 0$. For any $\eta > 0$,*

$$\lim_{n \rightarrow \infty} \Pr\left(\sup_{t \in [0, T_0]} \frac{1}{n} W_t^{(n)}\left([C^{(n)}(t), F^{(n)}(t)]\right) > \eta\right) = 0.$$

Proof of Lemma EC.3. Under Assumption 1, every job has service time $\sigma \geq s_{\min} > 0$, so

$$W_t^{(n)}\left([C^{(n)}(t), F^{(n)}(t)]\right) \geq s_{\min} \Phi_t^{Q(n), \tau}\left([C^{(n)}(t), F^{(n)}(t)]\right).$$

Taking $\sup_{t \in [0, T_0]}$ and dividing by n , Lemma EC.8 implies

$$\sup_{t \in [0, T_0]} \frac{1}{n} \Phi_t^{Q(n), \tau}\left([C^{(n)}(t), F^{(n)}(t)]\right) \leq \frac{1}{s_{\min}} \sup_{t \in [0, T_0]} \frac{1}{n} W_t^{(n)}\left([C^{(n)}(t), F^{(n)}(t)]\right) \xrightarrow{P} 0,$$

which is precisely the claim. $\square$

We now verify the auxiliary results. The proofs are ordered so that each proof uses only results that have already been established: we begin with the three leaf lemmas, then build upward.

Proof of Lemma EC.4. Define $g(\tau) := \lambda \mathbb{E}[S \mathbf{1}_{\{T \leq \tau\}}]$ for $\tau \in [T_{\min}, T_{\max}]$. By Assumption 1, S is bounded, so dominated convergence shows g is continuous. For $\tau_1 < \tau_2$,

$$g(\tau_2) - g(\tau_1) = \lambda \mathbb{E}[S \mathbf{1}_{\{\tau_1 < T \leq \tau_2\}}] \geq \lambda s_{\min} \int_{\tau_1}^{\tau_2} f_T(t) dt > 0,$$

where the positivity uses the full-support condition $f_T(t) > 0$ on $(T_{\min}, T_{\max})$. Hence g is strictly increasing. Since $g(T_{\min}) = 0$ (as $\Pr(T \leq T_{\min}) = 0$ from Assumption 1) and $g(T_{\max}) = \lambda \mathbb{E}[S] > N$ (the last equality using $\Pr(T \leq T_{\max}) = 1$, and the inequality by Assumption 2), the intermediate value theorem yields a unique $\bar{\tau} \in (T_{\min}, T_{\max})$. $\square$

Proof of Lemma EC.5. Define $W_0(I) := \int_{\mathcal{R}_0} \sigma \mathbf{1}_{\{\tau \in I\}} q_0(d\sigma, d\tau)$ for $I \in \mathcal{B}(\mathbb{R}_+)$. For any bounded continuous g , the map $(\sigma, \tau) \mapsto \sigma g(\tau)$ is bounded and continuous on $\mathcal{R}_0$, so weak convergence $n^{-1} q_0^{(n)} \Rightarrow q_0$ (Assumption 3) gives

$$\langle n^{-1} W_0^{(n)}, g \rangle = \int_{\mathcal{R}_0} \sigma g(\tau) (n^{-1} q_0^{(n)})(d\sigma, d\tau) \rightarrow \int_{\mathcal{R}_0} \sigma g(\tau) q_0(d\sigma, d\tau) = \langle W_0, g \rangle.$$

Hence $n^{-1} W_0^{(n)} \Rightarrow W_0$. The bounds $s_{\min} q_0^\tau(I) \leq W_0(I) \leq s_{\max} q_0^\tau(I)$ and Assumption 3 yield $W_0([0, F(0)]) = 0$ and $W_0([0, x]) > 0$ for $x > F(0)$. $\square$

Proof of Lemma EC.6. Fix $x > 0$. By Assumption 3, for each fixed $\xi > 0$,

$$\frac{1}{n} \tilde{G}_0^{(n)}(\xi) = \frac{|I_0^{(n)}|}{n} \cdot \frac{1}{|I_0^{(n)}|} \tilde{G}_0^{(n)}(\xi) \xrightarrow{P} \kappa_0 \mathbb{E}[S \mathbf{1}_{\{0 \leq T - S < \xi\}}]$$

by Slutsky's theorem. Under Assumption 1, $\Pr(0 \leq T - S < \xi) \rightarrow \Pr(T = S) = 0$ as $\xi \downarrow 0$, so $\kappa_0 \mathbb{E}[S \mathbf{1}_{\{0 \leq T - S < \xi\}}] \rightarrow 0$. Choose ξ small enough that this limit is below $x/2$. Then $\Pr(n^{-1} \tilde{G}_0^{(n)}(\xi) > x) \rightarrow 0$ by convergence in probability, and the double limit follows. $\square$

Proof of Lemma EC.7. By Lemma EC.4, $\bar{\tau}$ exists and is unique. Fix $T_0 > 0$ and $\epsilon > 0$. Choose $\xi \in (0, \epsilon/2)$ and decompose

$$\Pr\left(\sup_{t \in [0, T_0]} F^{(n)}(t) > \bar{\tau} + \epsilon\right) \leq \Pr\left(\sup_{t \in [0, \xi]} F^{(n)}(t) > \bar{\tau} + \epsilon\right) + \Pr\left(\sup_{t \in [\xi, T_0]} F^{(n)}(t) > \bar{\tau} + \epsilon\right). \quad (\text{EC.6})$$

Step 1 (early times, $t \in [0, \xi]$). By Assumption 3, $F^{(n)}(0) \leq \bar{\tau} + \epsilon/2$ with probability tending to 1. On $\{F^{(n)}(0) \leq \bar{\tau} + \epsilon/2\} \cap \{\sup_{t \in [0, \xi]} F^{(n)}(t) > \bar{\tau} + \epsilon\}$, the frontier must jump upward, which requires an upward jump of the HOL time-to-deadline $c^{(n)}$. There exists $\nu^{(n)} \in [0, \xi]$ with $c^{(n)}(\nu^{(n)}) > \bar{\tau} + \epsilon/2$. At this time, the queue has no waiting mass with time-to-deadline below $\bar{\tau} + \epsilon/2$, so all initially present mass with times-to-deadline in $(\xi, \bar{\tau} + \epsilon/2]$ must have left the queue by time ξ . The workload from these requests satisfies

$$\frac{1}{n}U^{(n)}(0, \xi] \geq \frac{1}{n}\left(W_0^{(n)}((\xi, \bar{\tau} + \epsilon/2]) - \tilde{G}_0^{(n)}(\xi)\right),$$

where the subtracted term accounts for initially present requests whose slack $T - S < \xi$ is too small to allow completion. By Lemma EC.2, $n^{-1}U^{(n)}(0, \xi] \leq N\xi$, so

$$N\xi \geq \frac{1}{n}\left(W_0^{(n)}((\xi, \bar{\tau} + \epsilon/2]) - \tilde{G}_0^{(n)}(\xi)\right). \quad (\text{EC.7})$$

By Lemma EC.5, $W_0((\xi, \bar{\tau} + \epsilon/2]) \rightarrow W_0([0, \bar{\tau} + \epsilon/2]) > 0$ as $\xi \downarrow 0$. Choose $x \in (0, W_0([0, \bar{\tau} + \epsilon/2])/4]$ and $\xi_1 > 0$ such that $W_0((\xi_1, \bar{\tau} + \epsilon/2]) > 2N\xi_1 + 2x$. By Lemma EC.6, there exists $\xi_2 > 0$ such that $\limsup_{n \rightarrow \infty} \Pr(n^{-1}\tilde{G}_0^{(n)}(\xi) > x) \leq x$ for $\xi \in (0, \xi_2]$. Set $\xi := \min\{\xi_1, \xi_2, \epsilon/2\}$. Then with probability tending to 1, the right-hand side of (EC.7) exceeds $2N\xi + x > N\xi$, contradicting the left-hand side. Hence $\Pr(\sup_{t \in [0, \xi]} F^{(n)}(t) > \bar{\tau} + \epsilon) \rightarrow 0$.

Step 2 (later times, $t \in [\xi, T_0]$). On $\{\sup_{t \in [\xi, T_0]} F^{(n)}(t) > \bar{\tau} + \epsilon\}$, there exists $\nu^{(n)} \in (\xi, T_0]$ with $c^{(n)}(\nu^{(n)}) > \bar{\tau} + \epsilon/2$. For any $\eta \in (0, \xi)$, all requests arriving in $(\nu^{(n)} - \eta, \nu^{(n)})$ whose times-to-deadline at $\nu^{(n)}$ fall in $(0, \bar{\tau} + \epsilon/2]$ must have entered service by $\nu^{(n)}$. By Lemma EC.2, $n^{-1}U^{(n)}(\nu^{(n)} - \eta, \nu^{(n)}) \leq N\eta$, and by Remark EC.1,

$$\frac{1}{n} \sum_{i: a_i^{(n)} \in (\nu^{(n)} - \eta, \nu^{(n)})} S_i^{(n)} \mathbf{1}_{\{S_i^{(n)} + \eta \leq T_i^{(n)} < \bar{\tau} + \epsilon/2\}} \geq \lambda\eta \mathbb{E}[S \mathbf{1}_{\{S + \eta \leq T < \bar{\tau} + \epsilon/2\}}] - 2\tilde{\epsilon}_A(n)$$

with probability at least $1 - \tilde{\epsilon}_A(n)$. As $\eta \downarrow 0$, $\mathbb{E}[S \mathbf{1}_{\{S + \eta \leq T < \bar{\tau} + \epsilon/2\}}] \uparrow \mathbb{E}[S \mathbf{1}_{\{T < \bar{\tau} + \epsilon/2\}}] > N/\lambda$ (using $N = \lambda \mathbb{E}[S \mathbf{1}_{\{T \leq \bar{\tau}\}}]$). For small η , the lower bound exceeds $N\eta$, contradicting the capacity bound. Hence the second probability in (EC.6) also vanishes. $\square$

Proof of Lemma EC.8. Fix $t > 0$ and $\eta > 0$. When $C^{(n)}(t) = F^{(n)}(t)$ the claim is trivial; assume $C^{(n)}(t) < F^{(n)}(t)$. Define

$$\omega^{(n)}(t) := \sup\{s \in [0, t] : C^{(n)}(s) = F^{(n)}(s)\}.$$

Since $C^{(n)}$ and $F^{(n)}$ are càdlàg and both decrease at unit rate between jumps, we have $C^{(n)}(\omega^{(n)}(t)-) = F^{(n)}(\omega^{(n)}(t)-)$ while $C^{(n)}(\omega^{(n)}(t)) < F^{(n)}(\omega^{(n)}(t))$. This implies $F^{(n)}(\omega^{(n)}(t)-) = F^{(n)}(\omega^{(n)}(t))$ and there must be an arrival at time $\omega^{(n)}(t)$.

At time $\omega^{(n)}(t)$, any request that arrived before $\omega^{(n)}(t)$ and is still waiting has remaining time-to-deadline at least $C^{(n)}(\omega^{(n)}(t)-) = F^{(n)}(\omega^{(n)}(t))$. So at most one waiting request (the new arrival) has remaining time-to-deadline in $[C^{(n)}(\omega^{(n)}(t)), F^{(n)}(\omega^{(n)}(t))]$, giving

$$W_{\omega^{(n)}(t)}^{(n)}\left([C^{(n)}(\omega^{(n)}(t)), F^{(n)}(\omega^{(n)}(t))]\right) \leq s_{\max}.$$

Since $C^{(n)} < F^{(n)}$ on $[\omega^{(n)}(t), t]$, the frontier $F^{(n)}$ decreases linearly: $F^{(n)}(s) = F^{(n)}(\omega^{(n)}(t)) - (s - \omega^{(n)}(t))$. Accounting for workload from pre- $\omega^{(n)}(t)$ requests, new arrivals, and service completions:

$$W_t^{(n)}\left([C^{(n)}(t), F^{(n)}(t)]\right) \leq s_{\max} + A^{(n)}(\omega^{(n)}(t), t] - U^{(n)}(\omega^{(n)}(t), t],$$

where $A^{(n)}(s, t]$ denotes the total arriving workload in $(s, t]$ from requests that remain feasible at t and have remaining time-to-deadline below $F^{(n)}(t)$.

By Lemma EC.7, $F^{(n)}(t) \leq \bar{\tau} + \epsilon$ for all $t \in [0, T_0]$ with probability tending to 1. On this event,

$$A^{(n)}(\omega^{(n)}(t), t] \leq \sum_{i: a_i^{(n)} \in (\omega^{(n)}(t), t]} S_i^{(n)} \mathbf{1}_{\{T_i^{(n)} < \bar{\tau} + \epsilon\}}.$$

By Remark EC.1, with probability at least $1 - \varepsilon_A(n)$,

$$\frac{1}{n} A^{(n)}(\omega^{(n)}(t), t] \leq \lambda(t - \omega^{(n)}(t)) \mathbb{E}[S \mathbf{1}_{\{T \leq \bar{\tau} + \epsilon\}}] + 2\varepsilon_A(n).$$

Since $C^{(n)} < F^{(n)}$ on $[\omega^{(n)}(t), t]$, the queue is nonempty throughout, so by Lemma EC.2, $n^{-1}U^{(n)}(\omega^{(n)}(t), t] = N(t - \omega^{(n)}(t))$. Combining:

$$\frac{1}{n} W_t^{(n)}\left([C^{(n)}(t), F^{(n)}(t)]\right) \leq \frac{s_{\max}}{n} + (t - \omega^{(n)}(t))(\lambda \mathbb{E}[S \mathbf{1}_{\{T \leq \bar{\tau} + \epsilon\}}] - N) + 2\varepsilon_A(n).$$

Since $N = \lambda \mathbb{E}[S \mathbf{1}_{\{T \leq \bar{\tau}\}}]$, the coefficient $\lambda \mathbb{E}[S \mathbf{1}_{\{T \leq \bar{\tau} + \epsilon\}}] - N$ can be made arbitrarily small by choosing ϵ small. For large n , each of the three terms is $< \eta/3$, so the supremum over $t \in [0, T_0]$ is at most η with probability tending to 1. $\square$

EC.1.3. Proof of Theorem 1 (Steady-state threshold and performance)

The proof requires the following single-type characterization, whose proof is given at the end of this subsection.

PROPOSITION EC.1 (Single-type case). *Consider a single-type system with arrival rate λ , service-time density f_S , and clinical-window density f_T on $\mathcal{R}_0$. Under the EDF policy in overload, the case-loss fraction is $\Pr(S > \tau^*)$ and the expected waiting time is $\mathbb{E}[(T - \max\{S, \tau^*\})^+]$, where $\tau^* \in (s_{\min}, s_{\max})$ is the unique solution of*

$$N = \lambda \int_0^{\tau^*} \sigma f_S(\sigma) d\sigma.$$

Proof of Theorem 1. Under the pooled-queue EDF policy, both early and late engagers share the same queue with arrival density $f = \alpha f^E + (1 - \alpha)f^L$. Since the steady-state PDE (derived in the proof of Proposition EC.1) is linear in the arrival density, the pooled system satisfies the same capacity-balance equation as a single-type system with density f . By Proposition EC.1, the threshold $\tau^* \in (s_{\min}, s_{\max})$ is the unique solution of $N = \lambda \int_0^{\tau^*} \sigma f_S(\sigma) d\sigma$, and a request with service time S completes service if and only if $S \leq \tau^*$.

The case-loss fractions follow by conditioning on type: an early engager is lost if $S^E > \tau^*$, a late engager if $S^L > \tau^*$.

For waiting times, a request with service time $S \leq \tau^*$ and clinical window T waits until its remaining time-to-deadline falls to τ^* (if $T > \tau^*$) or enters service immediately (if $T \leq \tau^*$). A request with $S > \tau^*$ waits $T - S$ until case loss. In both cases, the waiting time is $W = (T - \max\{S, \tau^*\})^+$. The type-specific expected waiting times follow by conditioning on type: $\mathbb{E}[(T^k - \max\{S^k, \tau^*\})^+]$ for $k \in \{E, L\}$. $\square$

We now prove the single-type characterization.

Proof of Proposition EC.1. In steady state, the fluid model density $g(\sigma, \tau)$ satisfies $\partial g / \partial t = 0$. The scheduling vector field under EDF is $\vec{u}(\sigma, \tau) = -(r(\sigma, \tau), 1)$ with $r(\sigma, \tau) = \mathbf{1}\{\tau < \tau^* \text{ and } \tau \geq \sigma\}$. Mass conservation over any region $\mathcal{R} \subset \mathcal{R}_0$ gives

$$\frac{d}{dt} \iint_{\mathcal{R}} g d\sigma d\tau = \lambda \iint_{\mathcal{R}} f d\sigma d\tau - \iint_{\mathcal{R}} \nabla \cdot (g\vec{u}) d\sigma d\tau,$$

where the last equality uses the divergence theorem. Since this holds for every $\mathcal{R}$, the steady-state PDE is

$$\frac{\partial(rg)}{\partial\sigma} + \frac{\partial g}{\partial\tau} + \lambda f = 0, \quad (\text{EC.8})$$

which splits into two regions:

$$\frac{\partial g}{\partial\tau} + \lambda f = 0, \quad \tau \geq \max\{\tau^*, \sigma\}, \quad (\text{EC.9})$$

$$\frac{\partial g}{\partial\sigma} + \frac{\partial g}{\partial\tau} + \lambda f = 0, \quad 0 < \sigma < \tau < \tau^*. \quad (\text{EC.10})$$

Solving the waiting region. Integrating (EC.9) in τ from τ to ∞ and using $g(\sigma, y) \rightarrow 0$ as $y \rightarrow \infty$ (by compact clinical-window support),

$$g(\sigma, \tau) = \lambda \int_{\tau}^{\infty} f(\sigma, x) dx, \quad \tau \geq \max\{\tau^*, \sigma\}. \quad (\text{EC.11})$$

Solving the service region. Along the characteristics $x \mapsto (\sigma + x, \tau + x)$ of (EC.10), $\frac{d}{dx} g(\sigma + x, \tau + x) = -\lambda f(\sigma + x, \tau + x)$. Setting $x_0 = \tau^* - \tau$ and using (EC.11) as the boundary condition at $\tau = \tau^*$:

$$g(\sigma, \tau) = \lambda \int_{\tau^*}^{\infty} f(\sigma + \tau^* - \tau, x) dx + \lambda \int_0^{\tau^* - \tau} f(\sigma + x, \tau + x) dx, \quad 0 < \sigma < \tau < \tau^*. \quad (\text{EC.12})$$

Capacity-balance equation. In overload, steady-state service rate equals capacity:

$$N = \int_0^{\tau^*} \int_0^{\tau} g(\sigma, \tau) d\sigma d\tau =: I_1 + I_2,$$

where I_1 and I_2 correspond to the two terms in (EC.12). For I_1 , substituting $\omega = \sigma + \tau^* - \tau$ and exchanging the order of integration gives

$$I_1 = \lambda \int_0^{\tau^*} \sigma \int_{\tau^*}^{\infty} f(\sigma, \tau) d\tau d\sigma.$$

For I_2 , substituting $(\omega, \nu) = (\sigma + x, \tau + x)$ and exchanging the order:

$$I_2 = \lambda \int_0^{\tau^*} \sigma \int_{\sigma}^{\tau^*} f(\sigma, \tau) d\tau d\sigma.$$

Adding and using $f_S(\sigma) = \int_{\sigma}^{\infty} f(\sigma, \tau) d\tau$:

$$N = \lambda \int_0^{\tau^*} \sigma f_S(\sigma) d\sigma.$$

Existence and uniqueness. Recall that $h(x) := \int_0^x \sigma f_S(\sigma) d\sigma$. Under Assumption 1, λh is continuous and strictly increasing on $[s_{\min}, s_{\max}]$ with $\lambda h(s_{\min}) = 0 < N < \lambda \mathbb{E}[S] = \lambda h(s_{\max})$. By the intermediate value theorem, $\tau^* \in (s_{\min}, s_{\max})$ exists and is unique.

Case-loss fraction. In steady state, mass exits via case loss along $\tau = \sigma$. The case-loss count per unit time is $\int_{\tau^*}^{\infty} g(\sigma, \sigma) d\sigma$. By (EC.11), $g(\sigma, \sigma) = \lambda \int_{\sigma}^{\infty} f(\sigma, x) dx = \lambda f_S(\sigma)$ for $\sigma \geq \tau^*$. The case-loss fraction is

$$\frac{1}{\lambda} \int_{\tau^*}^{\infty} \lambda f_S(\sigma) d\sigma = \Pr(S > \tau^*).$$

Waiting time. A request with service time $S \leq \tau^*$ and clinical window T is completed. If $T > \tau^*$, the request waits $T - \tau^*$ until its remaining time-to-deadline reaches τ^* . If $T \leq \tau^*$, service starts immediately. A request with $S > \tau^*$ waits $T - S$ until case loss. The overall waiting time is $W = (T - \max\{S, \tau^*\})^+$. $\square$

EC.2. Proofs of Results in Section 5

Fix a reservation fraction $\gamma \in [0, 1]$. The system operates as two independent EDF queues: a dedicated queue for early engagers with capacity γN , and a dedicated queue for late engagers with capacity $(1 - \gamma)N$.

EC.2.1. Proof of Proposition 1

Proof of Proposition 1. Consider the dedicated early-engager queue with arrival rate λ^E , service-time density f_S^E , and capacity γN . Two regimes arise.

- (i) If $\gamma N < \lambda^E \mathbb{E}[S^E]$, the queue is overloaded. By Theorem 1 applied to this single-class system, the EDF serving threshold $\tau^{E*}(\gamma)$ is the unique solution to

$$\gamma N = \lambda^E \int_0^{\tau^{E*}(\gamma)} \sigma f_S^E(\sigma) d\sigma = \lambda^E h^E(\tau^{E*}(\gamma)).$$

Therefore $\tau^{E*}(\gamma) = (h^E)^{-1}(\gamma N / \lambda^E)$. This regime corresponds to $0 \leq \gamma < \gamma^{max}$.

- (ii) If $\gamma N \geq \lambda^E \mathbb{E}[S^E]$, the capacity suffices for all early-engager workload, so no EDF cutoff binds in steady state and we set $\tau^{E*}(\gamma) = T_{max}^E$. This regime corresponds to $\gamma \geq \gamma^{max}$.

Combining (i)–(ii) yields the stated piecewise definition of $\tau^{E*}(\gamma)$.

The same reasoning applied to the dedicated late-engager queue with arrival rate λ^L , service-time density f_S^L , and capacity $(1 - \gamma)N$ yields the piecewise definition of $\tau^{L*}(\gamma)$:

- (i) In overload ($\gamma > \gamma^{min}$): $\tau^{L*}(\gamma) = (h^L)^{-1}((1 - \gamma)N / \lambda^L)$.
(ii) In underload ($\gamma \leq \gamma^{min}$): $\tau^{L*}(\gamma) = T_{max}^L$.

In a single-class EDF system at steady state with threshold τ^* , a request is served if and only if its service-time requirement does not exceed τ^* . The case-loss fraction is therefore $\Pr(S > \tau^*)$. Applying this to each independent queue gives the stated case-loss fractions and, after multiplication by the arrival rate, the case-loss counts per unit time. $\square$

EC.2.2. Proof of Lemma 2

Proof of Lemma 2. Step 1: Monotonicity of thresholds. For $\gamma \in I$, both queues are overloaded and the thresholds are uniquely determined by

$$\gamma N = \lambda^E h^E(\tau^{E*}(\gamma)) \quad \text{and} \quad (1 - \gamma)N = \lambda^L h^L(\tau^{L*}(\gamma)).$$

Differentiating implicitly:

$$\frac{d}{d\gamma} \tau^{E*}(\gamma) = \frac{N}{\lambda^E \tau^{E*}(\gamma) f_S^E(\tau^{E*}(\gamma))} > 0, \quad \frac{d}{d\gamma} \tau^{L*}(\gamma) = -\frac{N}{\lambda^L \tau^{L*}(\gamma) f_S^L(\tau^{L*}(\gamma))} < 0.$$

Step 2: Derivative of $C(\gamma)$. By Proposition 1,

$$C(\gamma) = c^E \lambda^E \left[1 - F_S^E(\tau^{E*}(\gamma)) \right] + c^L \lambda^L \left[1 - F_S^L(\tau^{L*}(\gamma)) \right].$$

Differentiating and substituting the threshold derivatives from Step 1:

$$\begin{aligned} C'(\gamma) &= -c^E \lambda^E f_S^E(\tau^{E*}(\gamma)) \frac{d}{d\gamma} \tau^{E*}(\gamma) - c^L \lambda^L f_S^L(\tau^{L*}(\gamma)) \frac{d}{d\gamma} \tau^{L*}(\gamma) \\ &= -\frac{c^E N}{\tau^{E*}(\gamma)} + \frac{c^L N}{\tau^{L*}(\gamma)} = N \left(\frac{c^L}{\tau^{L*}(\gamma)} - \frac{c^E}{\tau^{E*}(\gamma)} \right). \end{aligned}$$

Step 3: Strict convexity and characterization of γ^u . Since $\tau^{E*}(\gamma)$ is strictly increasing and $\tau^{L*}(\gamma)$ is strictly decreasing on I (Step 1), $c^E / \tau^{E*}(\gamma)$ is strictly decreasing and $c^L / \tau^{L*}(\gamma)$ is strictly increasing. Hence $C'(\gamma)$ is continuous and strictly increasing on I , so C is strictly convex with a unique minimizer $\gamma^u \in I$.

If $C'(\gamma) < 0$ for all $\gamma \in I$ —equivalently, $c^E / \tau^{E*}(\gamma) > c^L / \tau^{L*}(\gamma)$ for all $\gamma \in I$ —then C is strictly decreasing and $\gamma^u = \gamma^{max}$. If $C'(\gamma) > 0$ for all $\gamma \in I$ —equivalently, $c^L / \tau^{L*}(\gamma) > c^E / \tau^{E*}(\gamma)$ for all $\gamma \in I$ —then C is strictly increasing and $\gamma^u = \gamma^{min}$. Otherwise, by the intermediate value theorem, $C'(\gamma^u) = 0$ at a unique interior point, giving $c^E / \tau^{E*}(\gamma^u) = c^L / \tau^{L*}(\gamma^u)$. $\square$

EC.2.3. Proof of Theorem 2

We first establish that the optimal reservation fraction lies in I .

LEMMA EC.9. *The optimal solution γ^* to (6) satisfies $\gamma^* \in I = [\gamma^{\min}, \gamma^{\max}]$.*

Proof of Lemma EC.9. By Proposition 1:

(i) For $\gamma \geq \gamma^{\max}$, the early-engager queue is underloaded: $A^E(\gamma) = 0$.

(ii) For $\gamma \leq \gamma^{\min}$, the late-engager queue is underloaded: $A^L(\gamma) = 0$.

We prove $\gamma^* \leq \gamma^{\max}$ by contradiction. Suppose $\gamma^* > \gamma^{\max}$. Let $\tilde{\gamma} := \gamma^* - \epsilon$ for sufficiently small $\epsilon > 0$, so that $\tilde{\gamma} > \gamma^{\max}$ and $A^E(\gamma^*) = A^E(\tilde{\gamma}) = 0$. Since the system is overloaded (Assumption 2) and $\gamma^* N > \lambda^E \mathbb{E}[S^E]$, we have $(1 - \gamma^*)N < \lambda^L \mathbb{E}[S^L]$, so the late-engager queue is overloaded. For sufficiently small ϵ , $(1 - \tilde{\gamma})N < \lambda^L \mathbb{E}[S^L]$ as well. By Proposition 1,

$$\tau^{L*}(\gamma^*) = (h^L)^{-1} \left(\frac{(1 - \gamma^*)N}{\lambda^L} \right), \quad \tau^{L*}(\tilde{\gamma}) = (h^L)^{-1} \left(\frac{(1 - \tilde{\gamma})N}{\lambda^L} \right).$$

Since $\gamma^* > \tilde{\gamma}$ and $(h^L)^{-1}$ is strictly increasing, $\tau^{L*}(\gamma^*) < \tau^{L*}(\tilde{\gamma})$, whence

$$A^L(\gamma^*) = \lambda^L \Pr(S^L > \tau^{L*}(\gamma^*)) > \lambda^L \Pr(S^L > \tau^{L*}(\tilde{\gamma})) = A^L(\tilde{\gamma}).$$

Therefore $C(\gamma^*) > C(\tilde{\gamma})$. Moreover,

$$\left| \frac{A^E(\tilde{\gamma})}{\lambda^E} - \frac{A^L(\tilde{\gamma})}{\lambda^L} \right| = \frac{A^L(\tilde{\gamma})}{\lambda^L} < \frac{A^L(\gamma^*)}{\lambda^L} = \left| \frac{A^E(\gamma^*)}{\lambda^E} - \frac{A^L(\gamma^*)}{\lambda^L} \right| \leq \Delta,$$

so $\tilde{\gamma}$ is feasible with strictly lower cost, contradicting the optimality of γ^* .

The proof that $\gamma^* \geq \gamma^{\min}$ is symmetric: suppose $\gamma^* < \gamma^{\min}$, set $\hat{\gamma} = \gamma^* + \epsilon$, and swap the roles of the two queues to reach the same contradiction. $\square$

Proof of Theorem 2. By Lemma EC.9, any optimal γ^* lies in I .

Step 1: Monotonicity of $\phi(\gamma)$. The signed fairness gap satisfies

$$\phi(\gamma) = \Pr(S^E > \tau^{E*}(\gamma)) - \Pr(S^L > \tau^{L*}(\gamma)) = F_S^L(\tau^{L*}(\gamma)) - F_S^E(\tau^{E*}(\gamma)).$$

Using the threshold derivatives from the proof of Lemma 2:

$$\phi'(\gamma) = f_S^L(\tau^{L*}(\gamma)) \frac{d}{d\gamma} \tau^{L*}(\gamma) - f_S^E(\tau^{E*}(\gamma)) \frac{d}{d\gamma} \tau^{E*}(\gamma) = -\frac{N}{\lambda^L \tau^{L*}(\gamma)} - \frac{N}{\lambda^E \tau^{E*}(\gamma)} < 0.$$

Thus ϕ is strictly decreasing and continuous on I . The constraint $|\phi(\gamma)| \leq \Delta$ is equivalent to $-\Delta \leq \phi(\gamma) \leq \Delta$.

If $\phi(\gamma) = \Delta$ has a solution in I , it is unique; call it $\underline{\gamma}$. Otherwise $\phi(\gamma) < \Delta$ for all $\gamma \in I$ and we set $\underline{\gamma} := \gamma^{\min}$. Similarly, if $\phi(\gamma) = -\Delta$ has a solution in I , it is unique; call it $\bar{\gamma}$. Otherwise set $\bar{\gamma} := \gamma^{\max}$.

Step 2: Projection onto the feasible set. Since (6) is feasible, there exists $\gamma \in I$ with $|\phi(\gamma)| \leq \Delta$, which implies $\underline{\gamma} \leq \bar{\gamma}$. The feasible set is

$$\mathcal{F} = \{\gamma \in I : |\phi(\gamma)| \leq \Delta\} = [\underline{\gamma}, \bar{\gamma}].$$

By Lemma 2, C is strictly convex on $I \supseteq \mathcal{F}$ with unique minimizer γ^u . The minimizer of C over $\mathcal{F}$ is therefore the projection of γ^u onto $\mathcal{F}$:

- (1) If $\gamma^u \in [\underline{\gamma}, \bar{\gamma}]$, then $\gamma^* = \gamma^u$.
- (2) If $\gamma^u < \underline{\gamma}$, then C is strictly decreasing on $(-\infty, \gamma^u)$ and hence strictly increasing on $\mathcal{F}$, so $\gamma^* = \underline{\gamma}$.
- (3) If $\gamma^u > \bar{\gamma}$, then C is strictly decreasing on $\mathcal{F}$, so $\gamma^* = \bar{\gamma}$.

□

EC.2.4. Proof of Corollary 1

Proof of Corollary 1. Consider the dedicated early-engager queue, which operates as a single-class EDF system with arrival rate λ^E , service-time distribution f_S^E , and capacity γN . For $\gamma \in (\gamma^{min}, \gamma^{max})$, the early-engager queue is overloaded. By Proposition 1, the steady-state EDF serving threshold is the unique solution to

$$\gamma N = \lambda^E h^E(\tau^{E*}(\gamma)) = \lambda^E \int_0^{\tau^{E*}(\gamma)} \sigma f_S^E(\sigma) d\sigma.$$

The right-hand side is a function of f_S^E and the ratio $\gamma N / \lambda^E$ only; neither the clinical-window distribution f_T^E nor the information-lead-time distribution f_I^E (the density of $I^E = e^E - r^E$) appears. Since h^E is continuous and strictly increasing on the support of f_S^E (Assumption 1), $\tau^{E*}(\gamma)$ is uniquely determined by f_S^E and $\gamma N / \lambda^E$. The case-loss fraction $\Pr(S^E > \tau^{E*}(\gamma)) = 1 - F_S^E(\tau^{E*}(\gamma))$ depends on $\tau^{E*}(\gamma)$ and F_S^E only, so it is also independent of f_T^E and f_I^E .

The same argument applies to the dedicated late-engager queue with arrival rate λ^L , service-time distribution f_S^L , and capacity $(1 - \gamma)N$: the threshold $\tau^{L*}(\gamma)$ and the case-loss fraction $\Pr(S^L > \tau^{L*}(\gamma))$ are determined by f_S^L and $(1 - \gamma)N / \lambda^L$, with no dependence on f_T^L or f_I^L .

By definition, $C(\gamma) = c^E \lambda^E \Pr(S^E > \tau^{E*}(\gamma)) + c^L \lambda^L \Pr(S^L > \tau^{L*}(\gamma))$ and $\phi(\gamma) = \Pr(S^E > \tau^{E*}(\gamma)) - \Pr(S^L > \tau^{L*}(\gamma))$. By Part (1), both expressions depend on γ only through the service-time distributions and capacity ratios. Since γ^* is determined by $C(\gamma)$, $\phi(\gamma)$, and the disparity tolerance Δ (Theorem 2), it is also independent of the clinical-window distributions (f_T^E, f_T^L) and the information-lead-time distributions (f_I^E, f_I^L) . □

EC.3. Proofs of Results in Section 6

EC.3.1. Proof of Theorem 3

Proof of Theorem 3. By Lemma EC.9, any optimal solution to (10) lies in $I = [\gamma^{min}, \gamma^{max}]$. The argument depends only on the case-loss counts per unit time $A^E(\gamma)$ and $A^L(\gamma)$, which are unchanged by the linear term $v\gamma N$: shifting all of γ capacity to one type and showing a strictly dominating feasible point proceeds exactly as in the proof of Lemma EC.9.

Step 1: Strict convexity of $\tilde{C}(\gamma)$. The extended cost is $\tilde{C}(\gamma) = C(\gamma) - v\gamma N$, so

$$\tilde{C}'(\gamma) = C'(\gamma) - vN \quad \text{and} \quad \tilde{C}''(\gamma) = C''(\gamma).$$

By Lemma 2, $C(\gamma)$ is strictly convex on I with derivative

$$C'(\gamma) = N \left(\frac{c^L}{\tau^{L*}(\gamma)} - \frac{c^E}{\tau^{E*}(\gamma)} \right),$$

and $C'(\gamma)$ is continuous and strictly increasing on I . Therefore $C''(\gamma) > 0$ for all $\gamma \in I$, and

$$\tilde{C}''(\gamma) = C''(\gamma) > 0, \quad \gamma \in I.$$

The function $\tilde{C}$ is strictly convex on I and admits a unique minimizer $\tilde{\gamma}^u \in I$.

Step 2: Characterization of $\tilde{\gamma}^u$. Substituting the expression for $C'(\gamma)$:

$$\tilde{C}'(\gamma) = N \left(\frac{c^L}{\tau^{L*}(\gamma)} - \frac{c^E}{\tau^{E*}(\gamma)} - v \right). \quad (\text{EC.13})$$

Three cases arise:

- (1) If $\tilde{C}'(\gamma^{\min}) \geq 0$, equivalently $c^L/\tau^{L*}(\gamma^{\min}) - c^E/\tau^{E*}(\gamma^{\min}) \geq v$, then $\tilde{C}$ is non-decreasing on I and $\tilde{\gamma}^u = \gamma^{\min}$.
- (2) If $\tilde{C}'(\gamma^{\max}) \leq 0$, equivalently $c^L/\tau^{L*}(\gamma^{\max}) - c^E/\tau^{E*}(\gamma^{\max}) \leq v$, then $\tilde{C}$ is non-increasing on I and $\tilde{\gamma}^u = \gamma^{\max}$.
- (3) Otherwise, by the intermediate value theorem and strict monotonicity of $\tilde{C}'$, there exists a unique $\tilde{\gamma}^u \in (\gamma^{\min}, \gamma^{\max})$ satisfying $\tilde{C}'(\tilde{\gamma}^u) = 0$, i.e.,

$$\frac{c^L}{\tau^{L*}(\tilde{\gamma}^u)} - \frac{c^E}{\tau^{E*}(\tilde{\gamma}^u)} = v.$$

This is the shifted first-order condition: at the baseline optimum γ^u , the index gap is zero; at the extended optimum $\tilde{\gamma}^u$, it equals v .

Step 3: $\tilde{\gamma}^u \geq \gamma^u$. We consider three cases based on the position of γ^u in I .

- (a) If $\gamma^u = \gamma^{\min}$, then $\tilde{\gamma}^u \geq \gamma^{\min} = \gamma^u$ trivially since $\tilde{\gamma}^u \in I$.
- (b) If $\gamma^u \in (\gamma^{\min}, \gamma^{\max})$, then $C'(\gamma^u) = 0$, so

$$\tilde{C}'(\gamma^u) = C'(\gamma^u) - vN = -vN \leq 0.$$

Since $\tilde{C}'$ is strictly increasing and $\tilde{C}'(\gamma^u) \leq 0$, the minimizer satisfies $\tilde{\gamma}^u \geq \gamma^u$. When $v > 0$, $\tilde{C}'(\gamma^u) = -vN < 0$, so $\tilde{\gamma}^u > \gamma^u$ strictly.

- (c) If $\gamma^u = \gamma^{\max}$, then $C'(\gamma^{\max}) \leq 0$ (since C is non-increasing up to $\gamma^{\max}$), so $\tilde{C}'(\gamma^{\max}) = C'(\gamma^{\max}) - vN \leq 0$. Hence $\tilde{C}$ is non-increasing on all of I , giving $\tilde{\gamma}^u = \gamma^{\max} = \gamma^u$.

In all cases, $\tilde{\gamma}^u \geq \gamma^u$, with strict inequality when $v > 0$ and $\gamma^u \in (\gamma^{\min}, \gamma^{\max})$.

Step 4: Projection onto the feasible set. The equity constraint $|\phi(\gamma)| \leq \Delta$ depends only on the case-loss fractions $A^E(\gamma)/\lambda^E$ and $A^L(\gamma)/\lambda^L$, which are determined by $C(\gamma)$ and are independent of v . The signed

fairness gap $\phi(\gamma) = A^E(\gamma)/\lambda^E - A^L(\gamma)/\lambda^L$ and its monotonicity properties are unchanged from the proof of Theorem 2 (Step 3). The feasible set is therefore

$$\mathcal{F} = \{\gamma \in I : |\phi(\gamma)| \leq \Delta\} = I \cap [\underline{\gamma}, \bar{\gamma}],$$

with $\underline{\gamma}$ and $\bar{\gamma}$ identical to the baseline problem. Since $\tilde{C}$ is strictly convex on $I \supseteq \mathcal{F}$, the minimizer of $\tilde{C}$ over $\mathcal{F}$ is the projection of $\tilde{\gamma}^u$ onto $\mathcal{F}$:

- (1) If $\tilde{\gamma}^u \in [\underline{\gamma}, \bar{\gamma}]$, then $\tilde{\gamma}^* = \tilde{\gamma}^u$.
- (2) If $\tilde{\gamma}^u < \underline{\gamma}$, then $\tilde{C}$ is strictly increasing on $\mathcal{F}$, so $\tilde{\gamma}^* = \underline{\gamma}$.
- (3) If $\tilde{\gamma}^u > \bar{\gamma}$, then $\tilde{C}$ is strictly decreasing on $\mathcal{F}$, so $\tilde{\gamma}^* = \bar{\gamma}$.

□

EC.4. Parameters, Calibration, and Implementation Details

This appendix documents the parameters, the cost calibration in 2023/24 pounds sterling, the synthetic patient-stream generator, and the simulation protocol that support the case study of Section 7. Section EC.4.1 reports the baseline parameters, the cost calibration that anchors the normalized case-loss costs c^E and c^L , and the sensitivity-analysis ranges. Section EC.4.2 specifies the synthetic patient-stream generator and the discrete-event simulation protocol.

EC.4.1. Parameters

EC.4.1.1. Baseline parameters. Table EC.1 reports the baseline parameter values used throughout the case study. The weekly arrival rate λ and the weekly elective theatre slot count N are calibrated against the CUH 2024 ELCS aggregate statistics. The equity tolerance Δ is set to 20%, the upper bound recommended by the CUH maternity scheduling team; smaller values are reported in the sensitivity analysis (Table EC.3). The satisfaction value v is set to zero in the baseline, so the headline reservation γ^* minimizes case-loss cost only; the sensitivity to v is reported in Section 7.2.2. The case-loss costs c^E and c^L are normalized, with the underlying £calibration documented in Section EC.4.1.2 below. The application of simulation horizon H , warm-up length, replication count R , and master seed is documented in the simulation protocol of Section EC.4.2.3.

EC.4.1.2. Cost calibration in pounds sterling. A case loss in our model is an elective C-section patient who, owing to capacity shortfall in the elective theatre, is diverted to the unplanned/emergency C-section pathway. The economic quantity that anchors c^E and c^L is therefore the *incremental cost of the emergency pathway over the planned pathway*: extra theatre time on an unscheduled list, longer maternal length of stay (LoS), higher anesthetic complexity, higher neonatal escalation risk, and the indirect costs of opportunity loss on the emergency list. Because NHS payment systems and NHS Reference Costs aggregate within Healthcare Resource Group (HRG) codes and do not observe the early/late distinction directly, any

Table EC.1 Baseline calibrated parameters.

Parameter	Notation	Value
Total weekly arrival rate	λ	23.11 patients/week (2024)
Weekly elective theatre slots	N	18
Early-engager per-case-loss cost	c^E	1 ($\approx$ £500)
Late-engager per-case-loss cost	c^L	2 ($\approx$ £1,000)
Equity tolerance	Δ	20%
Satisfaction value	v	0
Synthetic-stream replications	R	100
Master seed	—	42
Simulation horizon	H	152 weeks, with 100-week warm-up
Measurement window	—	52 weeks ($\approx$ one calendar year)

£figure attached to the early/late split must be a reasoned argument from clinical-pathway differences rather than a direct billing read-out.

We anchor the headline differential on the NHS Reference Costs 2014/15 caesarean breakdown reproduced in NIHR HTA Vol. 23.9 Appendix 11 (Gallos et al. 2019). Activity-weighted national average unit costs are reported in Table EC.2 both in 2014/15 prices and uplifted to 2023/24 prices using the PSSRU NHS Cost Inflation Index (Jones and Burns 2024, cumulative factor ≈ 1.27). Two natural differentials emerge from the table: the activity-weighted non-elective minus elective differential is approximately £1,010 in 2023/24 prices, and the long-stay non-elective minus elective differential is approximately £1,300. A second corroborating set from Schaal et al. (2016) via NHS Maternity Statistics 2014/15 yields a differential of approximately £1,140 in 2023/24 prices. The NHS Payment Scheme 2023/25 (NHS England 2024a) preserves the relative gradient via the HRG codes NZ40A/B/C (planned) and NZ41A/B/C (non-elective).

Table EC.2 Activity-weighted NHS Reference Costs for caesarean section.

Category	Finished consultant episodes	Unit cost (2014/15)	Unit cost (2023/24)
Elective inpatient	5,745	£3,035	£3,856
Non-elective long stay	138,750	£4,060	£5,156
Non-elective short stay	20,987	£2,313	£2,937
All caesarean section, weighted	165,482	£3,803	£4,830

To translate the aggregate non-elective minus elective differential into the type-specific incremental costs c^E and c^L , we look at the detailed breakdown across the three clinical channels that drive the gradient between Category 1/2 emergency and Category 3 / semi-elective post-deadline conversion.

- *Length of stay.* A multi-center prospective UK cohort (Carvalho et al. 2023) identifies caesarean urgency as one of the strongest predictors of prolonged post-CS LoS; planned post-CS LoS is around 2.4 days, whereas Category 1/2 emergency presentations reach 4–5 days. At ward day costs of approximately £400–£500 (Jones and Burns 2024), a one-day differential moves $c^L - c^E$ by £400–£500.

- *Anesthesia mix.* General anesthesia (GA) is disproportionately used in Category 1 and rapid Category 2 emergencies because of the shorter decision-to-delivery interval (Hu et al. 2024); GA carries a higher per-case anesthetic resource cost and is associated with worse immediate neonatal outcomes. Late-engager case losses concentrate in the Category 1/2 subset and therefore in the GA-leaning anesthesia mix.
- *NICU admission gradient.* Australian data cited in the NICE NG192 evidence review (National Institute for Health and Care Excellence 2021a) report NICU admission rates of 9.8% versus 2.5% for Category 1 emergency CS versus non-Category 1 CS. At neonatal critical-care costs of approximately £1,100–£1,500 per day (Jones and Burns 2024), this gradient adds an expected-cost contribution of £200–£400 in favor of the late-engager differential.

Combining the three channels gives a plausible late-engager incremental cost of roughly £900–£1,400 over the planned pathway—in line with the long-stay non-elective minus elective figure of approximately £1,300 from Table EC.2—and a plausible early-engager incremental cost of roughly £400–£700.

The point estimates for the manuscript are therefore $c^E \approx £500$ (range £300–£700) and $c^L \approx £1,000$ (range £700–£1,400), with a literature-defensible plausibility range for c^L/c^E of [1.3, 2.5]. One normalized cost unit therefore corresponds to approximately £500 in 2023/24 prices, so $c^E = 1 \approx £500$ and $c^L = 2 \approx £1,000$. The paper’s normalized 2:1 ratio sits in the middle of the plausibility range, and the cost-ratio sensitivity reported in Section 7.3 sweeps this range plus an upper-extrapolation point at $c^L/c^E = 3.0$ for robustness. Two caveats deserve declaration: NHS Reference Costs aggregate within HRG codes, while the early-engager / late-engager distinction is an operational construct based on the request-to-earliest-eligibility gap, so the £figures should be read as informed approximations rather than patient-level invoice read-outs; and litigation and long-term morbidity costs are excluded from c^E and c^L , including them would shift the level of both upwards but is unlikely to change their ratio.

EC.4.1.3. Sensitivity ranges. Table EC.3 reports the ranges over which individual parameters are swept in the sensitivity and extension analyses of Sections 7.2.2 and 7.3. In each sweep, one parameter is varied while all others are held at their baseline values from Table EC.1.

EC.4.2. Synthetic Patient-Stream Generator

The case study draws synthetic patient streams from a deterministic-given-seed generator calibrated against CUH ELCS aggregate statistics. The aggregate calibration parameters are reported in Section EC.4.2.1; the per-patient generation algorithm and reproducibility protocol are documented in Section EC.4.2.2; and the discrete-event simulation that consumes the generator output is detailed in Section EC.4.2.3.

EC.4.2.1. Preparation. The aggregate calibration parameters for the 2024 in-sample and 2025 out-of-sample horizons are reported in Table EC.4. The 2024 horizon is the basis for the main results in Section 7; the 2025 horizon supports the out-of-sample validation in Section 7.1.

Table EC.3 Parameter ranges swept in the sensitivity and extension analyses.

Parameter	Range	Section
Equity tolerance, Δ	{5%, 10%, 15%, 20%}	Section 7.2.1
Satisfaction value, ν	{0, 0.1, 0.3, 0.5, 0.7, 1.0}	Section 7.2.2
Cost ratio, c^L/c^E (with $c^E + c^L = 3$)	{1.31, 2.00, 2.49, 3.00}	Section 7.3.1
Cohort cutoff (days) [†]	{20, 25, 30, 35, 40}	Section 7.3.2
Amplitude, a , in time-varying demand $\lambda_t = \lambda(1 + a \sin(2\pi t/52))$	{0.05, 0.10, 0.15, 0.20, 0.25}	Section 7.3.3
No-show probability, p_{ns}	{0, 0.02, 0.05}	Section 7.3.4
Weekly slot capacity, N	{16, 17, 18}	Section 7.3.5

[†] The five cutoff values produce realized early-engager shares $\alpha = \lambda^E/\lambda$ of 80.6%, 75.9%, 71.2%, 64.9%, and 59.2% respectively (see Section 7.3.2).

Table EC.4 Aggregate calibration parameters.

Parameter	Symbol	2024 (in-sample)	2025 (out-of-sample)
Weekly arrival rate	λ	23.11 patients/wk	25.37 patients/wk
Annual volume (52 weeks)	$\lambda \cdot 52$	1,202	1,319
Weekly elective slots	N	18	18
Service-mix probability	$\Pr(S=1)$	0.97	0.978
Service-mix probability	$\Pr(S=2)$	0.03	0.022
Indication mix	—	Tab. EC.5, col. 3	Tab. EC.5, col. 4
Type threshold	g	> 35 days $\Rightarrow$ early	> 35 days $\Rightarrow$ early

Table EC.5 Indication-level prevalences (2024 in-sample and 2025 out-of-sample), gestational-age clinical windows, and r-to-e gap 95% CIs.

#	Indication	2024 (%)	2025 (%)	GA_e	GA_d	r-to-e gap 95% CI (days)
1	Previous c-section	44.0	45.0	38+0 to 39+0	39+6 to 40+0	(30, 111)
2	Maternal request	20.0	21.0	39+0 (point)	40+0 (point)	(5, 80)
3	Breech presentation	9.5	9.0	39+0 (point)	39+6 to 40+0	(−7, 31)
4	Fetal causes	7.0	4.0	34+0 to 38+0	38+0 to 39+0	(−8, 67)
5	Maternal medical indication	6.0	8.0	34+0 to 38+0	38+0 to 39+0	(3, 91)
6	Multiple births	4.0	4.0	37+0 to 38+0	38+0 (point)	(9, 52)
7	Postdates	3.0	4.0	40+0 (point)	41+0 (point)	(−7, 32)
8	Placenta praevia	3.0	2.0	36+0 to 37+0	37+6 (point)	(1, 38)
9	Previous traumatic delivery	3.0	2.0	38+0 (point)	39+0 to 39+6	(30, 113)
10	Unstable lie	0.5	1.0	38+0 to 39+0	39+0 (point)	(−7, 33)
Total	—	100.0	100.0	—	—	—

Notes. Gestational ages are reported in the format “ $a+b$ ”, meaning $7a + b$ days from a clinical origin (last menstrual period). “ $a+b$ to $c+d$ ” denotes a closed integer-day interval $[7a + b, 7c + d]$ sampled uniformly; “ $a+b$ (point)” is a deterministic single value $7a + b$. GA_e is the earliest gestational age at which the surgery is clinically appropriate. GA_d is the latest gestational age at which the surgery should be performed: because a caesarean surgery (~ 40 – 60 min) is near-instantaneous on the gestational-age scale, a patient is clinically served as long as her surgery can *start* before GA_d .

Per-patient draws use the indication-level prevalence and clinical windows in Table EC.5. The clinical windows follow national clinical guidance. Specifically, NICE NG192 (National Institute for Health and Care Excellence 2021a) for previous caesarean and maternal request; RCOG Green-top 20b (Impey et al.

2017) for breech presentation; SMFM consensus (Spong et al. 2011) for fetal causes and maternal medical indication; NICE NG137 (National Institute for Health and Care Excellence 2019) for multiple births; NICE NG207 (National Institute for Health and Care Excellence 2021b) for postdates; RCOG Green-top 27a (Jauniaux et al. 2019) for placenta praevia. Two remaining indications (previous traumatic delivery, unstable lie) follow the collaborating Trust’s local protocol, which has no published guideline.

The r-to-e gap 95% CI for each indication is the indication-specific window over which the gap $g = e_i - r_i$ is sampled uniformly; negative lower bounds reflect indications detected at or after the earliest clinically appropriate gestational age and trigger the effective-eligibility rule of Section EC.4.2.2.

EC.4.2.2. Generation algorithm. The generator produces an ordered patient stream sorted by r_i , executing the following steps once per patient:

1. *Sample request times.* Draw $\{r_i\}$ as a homogeneous Poisson process on $[0, H]$ at rate λ per week, equivalently as i.i.d. exponential inter-arrival times, where $H = 152$ weeks.
2. *Sample indication.* Draw an indication independently from the categorical distribution in column 3 (2024) or column 4 (2025) of Table EC.5.
3. *Sample gestational ages.* Draw $GA_{e,i}$ and $GA_{d,i}$ (in days) independently and uniformly from the indication-specific windows in columns 5 and 6 of Table EC.5.
4. *Sample r-to-e gap.* Draw g_i uniformly over the indication-specific 95% CI in column 7 of Table EC.5.
5. *Translate to calendar time.* Set $e_i = r_i + g_i/7$ and $d_{\text{clinical},i} = e_i + (GA_{d,i} - GA_{e,i})/7$ (in weeks).
6. *Sample service requirement.* Draw $S_i \in \{1, 2\}$ independently from the year-specific service-mix probabilities in Table EC.4.
7. *Apply effective eligibility and feasibility filter.* Set $e_i \leftarrow \max\{e_i, r_i\}$ and verify $r_i \leq e_i \leq d_{\text{clinical},i}$; if the feasibility check fails, resample steps 2–6 for this r_i (capped at 100 resamples).
8. *Classify type.* A patient is classified as an early engager if $g_i > 35$ days, and as a late engager otherwise.
9. *Calibrated completion deadline.* Set the model’s completion deadline $d_i = d_{\text{clinical},i} + S_i$ (weeks).

A single integer master seed (42) controls all randomness. For replication r , the per-replication seed is $42 + 1000r$; from it the generator derives one sub-seed per random stream (arrivals, indication, gestational-age, r-to-e gap, service-time), so that the same master seed reproduces the patient stream byte-for-byte. The generator is run for $R = 100$ replications per calibration year, indexed $r = 1, \dots, R$.

EC.4.2.3. Simulation protocol. All results in Section 7 are obtained from a discrete-event simulator implementing non-preemptive within-class earliest-deadline-first scheduling. The simulator operates in weeks. The simulation parameters are summarized below.

- *Horizon and warm-up.* Each replication is simulated over a horizon of $H = 152$ weeks, with the first 100 weeks discarded as warm-up, so that all reported statistics are computed from approximately one calendar year of stationary operation.

- *Replications.* $R = 100$ independent replications per scenario, indexed $r = 1, \dots, 100$, with master seed 42 and per-replication seed $42 + 1000r$.
- *Common random numbers.* A single set of baseline patient traces is generated once per replication and reused across all policies, ensuring that observed differences between policies reflect the scheduling mechanism rather than sampling variability.
- *Stochastic rounding of slot allocations.* Fractional slot allocations (N^E, N^L) are implemented via per-week stochastic rounding: in each simulated week t , $N_t^E = \lfloor \gamma N \rfloor + \text{Bernoulli}(\gamma N - \lfloor \gamma N \rfloor)$ and $N_t^L = N - N_t^E$. Slots reserved for one cohort cannot serve the other. The long-run averages are $\mathbb{E}[N_t^E] = \gamma N$ and $\mathbb{E}[N_t^L] = (1 - \gamma)N$ exactly, with per-week sample paths fluctuating by one slot.
- *Optimal-reservation grid.* The optimal-reservation search is conducted on a discrete grid in γ with step size 0.01, selecting the feasible minimizer of $C(\gamma)$ subject to $|\phi(\gamma)| \leq \Delta$.
- *Confirmation-date rule.* Under current scheduling practice, the confirmation date b_i is fixed at the start of the patient's allocated surgery week, $b_i = \lfloor s_i \rfloor$, where s_i is the simulator's start-of-service time (both in weeks). This yields a confirmation lead time $s_i - b_i \in [0, 1)$ week per served patient, with a realized average of approximately 3.5 days (≈ 0.5 week) in the simulator. Under early confirmation (Section 7.2.3), $b_i = r_i$ for early engagers; the current scheduling practice still applies to late engagers.